The Heart Conjecture for Product Pure-State Channels
A Structural and Low-Dimensional Research Report

Jan Hązła and Ryan O'Donnell

August 2026

Abstract

We study four conjectured comparisons between product binary pure-state channels and matched random-coordinate reveal experiments. The report organizes the known results around a single emerging picture: Boolean parity interactions are the natural logical variables; near the uniform law all Walsh interactions decouple; near a perfect channel the Heart gap becomes a positive edge energy on the Boolean cube; and the odd part of a law is exactly a phase syndrome whose residual obstruction is information about a hidden activation set. We also give complete proofs of the three-bit uniform-marginal theorem and the antipodally even four-bit theorem.

The conjectures were made jointly by Jan Hązła and Ryan O'Donnell. All the theorems and proofs recorded here were obtained through prompting ChatGPT. Whenever a proof uses computer assistance, the exact finite algebraic statement certified by the computation is included, while the code and large raw output are omitted.

Contents

1 Conjectures and notation **3**

2 Main results and logical roadmap **4**

3 Common representations and analytic tools **6**

3.1 The Walsh-basis form of the channel 6

3.2 Pythagorean identities 7

3.3 The BKM Hessian and logarithmic-mean bound 7

3.4 Entropy and Mellin log-determinant formulas 8

3.5 A complete flagged-parity decomposition 8

4 Parity interactions and logical parity blocks **9**

4.1 Top-Walsh contraction 9

4.2 The one-bit inequality and its strictness 9

4.3 Two bits 10

4.4 Universal block-parity lifting 10

4.5 The nested-Walsh trace-moment family 11

5 Local geometry at the uniform and perfect-channel endpoints **12**

5.1 The full unpinched Hessian at the uniform law 12

5.2 The full-cube perfect-channel edge energy 14

5.3 A common neighborhood of the uniform perfect-channel corner 16

5.4 Neighborhoods of products of parity codes 16

6 Global reductions: number sectors, phase syndromes, and odd perturbations **17**
6.1 Replicated-type number-sector reduction . . . 17
6.2 The phase experiment . . . 18
6.3 The phase-syndrome identity . . . 18
6.4 Coefficientwise full-reveal contraction . . . 19
6.5 The isolated one-coordinate removal problem . . . 21

7 Proof of the three-bit theorem **21**
7.1 Removing the cubic interaction . . . 21
7.2 Even laws are four masses . . . 22
7.3 Compare two masses directly . . . 23
7.3.1 The exact pair formula . . . 23
7.3.2 The channel expansion has one moving root . . . 24
7.3.3 A universal weighted-mean estimate . . . 25
7.3.4 The constant coefficient is a convex edge energy . . . 26
7.4 KKT forces one small mass and three equal masses . . . 27
7.5 The remaining line is a two-by-two calculation . . . 28
7.5.1 Endpoint reserve, slope, and a chord . . . 29
7.5.2 Use the scalar toolkit . . . 31

8 Proof of the even four-bit theorem **31**
8.1 Klein-four reduction . . . 31
8.2 The Fano reserve . . . 32
8.2.1 1. A sharp reverse entropy inequality on two bits . . . 32
8.2.2 2. The sharp Fano-plane hierarchy . . . 34
8.3 Negative global-parity branch . . . 35
8.3.1 1. Golden–Thompson overlap entropy . . . 35
8.3.2 2. A sharper isoclinic MUB theorem . . . 36
8.3.3 3. Equality on the three Klein midpoints . . . 37
8.3.4 4. The one-bit excess is jointly convex . . . 37
8.3.5 5. Scalar Klein interpolation . . . 38
8.3.6 6. Payment by the actual classical branch . . . 38
8.4 Positive global-parity branch . . . 39
8.4.1 1. Result . . . 39
8.4.2 2. Exact swap testing curve . . . 40
8.4.3 3. Four breakpoint patterns . . . 41
8.4.4 4. Removing the fixed same-coset background . . . 42
8.4.5 5. A scalar background envelope . . . 42
8.4.6 6. The increasing-convex-order lemma . . . 43
8.4.7 7. The binary endpoint . . . 45
8.4.8 8. From the local Hessian to every swap amplitude . . . 46
8.4.9 9. Completion of the positive branch and of the even theorem . . . 46

1 Conjectures and notation

For coordinate i, let Φ_i be a binary pure-state channel with signal states $|\psi_+^{(i)}\rangle, |\psi_-^{(i)}\rangle$ of trace distance $s_i \in [0,1]$. We use the canonical realization

$$|\psi_x^{(i)}\rangle = \sqrt{\frac{1+\mu_i}{2}}|0\rangle + x\sqrt{\frac{1-\mu_i}{2}}|1\rangle, \qquad \mu_i = \sqrt{1-s_i^2}.$$

Write $\Phi_{\boldsymbol{s}} = \bigotimes_i \Phi_i$. Let $R \subseteq [n]$ be random, with $i \in R$ independently with probability s_i. For a law p on $\{\pm 1\}^n$, write p_R for its R-marginal and u_R for the uniform law on $\{\pm 1\}^R$.

For two laws p, q define the matched quantum and classical divergences

$$\mathcal{Q}_{\boldsymbol{s}}(p\|q) := D(\Phi_{\boldsymbol{s}}(p)\|\Phi_{\boldsymbol{s}}(q))\,, \qquad \mathcal{C}_{\boldsymbol{s}}(p\|q) := \mathbb{E}_R D(p_R\|q_R).$$

Conjecture 1 (Uniform-reference relative entropy; Conjecture 1 (Heart))**.** *For every n, every law p on $\{\pm 1\}^n$, and arbitrary channel parameters $\boldsymbol{s}$,*

$$\mathcal{Q}_{\boldsymbol{s}}(p\|u_n) \le \mathcal{C}_{\boldsymbol{s}}(p\|u_n).$$

Let $q = \bigotimes_i p_i$ be the product of the one-bit marginals of p.

Conjecture 2 (Product-reference relative entropy; Conjecture 2 (TC-Heart))**.** *For every p and arbitrary $\boldsymbol{s}$,*

$$\mathcal{Q}_{\boldsymbol{s}}(p\|q) \le \mathcal{C}_{\boldsymbol{s}}(p\|q).$$

When all one-bit marginals are uniform, Conjectures 1 and 2 coincide.

For $1 \le \alpha \le 2$, define the sandwiched trace moment

$$\mathsf{Q}_{\alpha,\boldsymbol{s}}(p\|q) := \mathrm{Tr}\Big[\Phi_{\boldsymbol{s}}(q)^{\frac{1-\alpha}{2\alpha}}\Phi_{\boldsymbol{s}}(p)\Phi_{\boldsymbol{s}}(q)^{\frac{1-\alpha}{2\alpha}}\Big]^{\alpha},$$

and the matched classical moment

$$\mathsf{B}_{\alpha,\boldsymbol{s}}(p\|q) := \mathbb{E}_R \sum_{x_R} p_R(x_R)^{\alpha} q_R(x_R)^{1-\alpha}.$$

Both sides equal 1 at $\alpha = 1$; their right derivatives are the two sides of the corresponding relative-entropy conjecture.

Conjecture 3 (Uniform-reference trace moments; Conjecture 3 (Box))**.** *For $1 \le \alpha \le 2$,*

$$\mathsf{Q}_{\alpha,\boldsymbol{s}}(p\|u_n) \le \mathsf{B}_{\alpha,\boldsymbol{s}}(p\|u_n).$$

Conjecture 4 (Product-reference trace moments; Conjecture 4 (TC-Box))**.** *For $1 \le \alpha \le 2$ and $q = \bigotimes_i p_i$,*

$$\mathsf{Q}_{\alpha,\boldsymbol{s}}(p\|q) \le \mathsf{B}_{\alpha,\boldsymbol{s}}(p\|q).$$

For density notation, write $p = f u_n$, so $\mathbb{E} f = 1$, and

$$\widehat{f}(S) = \mathbb{E}[f(X)\chi_S(X)], \qquad \chi_S(x) = \prod_{i\in S} x_i.$$

We abbreviate

$$\mathcal{C}_{\boldsymbol{s}}(f) := \mathcal{C}_{\boldsymbol{s}}(f u_n\|u_n), \qquad \mathcal{Q}_{\boldsymbol{s}}(f) := \mathcal{Q}_{\boldsymbol{s}}(f u_n\|u_n), \qquad \mathcal{G}_{\boldsymbol{s}}(f) := \mathcal{C}_{\boldsymbol{s}}(f) - \mathcal{Q}_{\boldsymbol{s}}(f).$$

For equal parameters we write $\mathcal{C}_s, \mathcal{Q}_s, \mathcal{G}_s$. Thus Heart is the assertion $\mathcal{G}_{\boldsymbol{s}}(f) \ge 0$. All logarithms are natural and $0\log 0 = 0$.

2 Main results and logical roadmap

The first results isolate individual Boolean interactions. A pure top-parity difference contracts by the exact probability that the reveal experiment sees every relevant coordinate. The flagged-parity decomposition then shows that a large physical block carrying only its parity is literally a logical one-bit pure-state channel, up to an input-independent flag.

Theorem 2.1 (A pure global-parity difference survives by at most the probability that the BEC reveals every coordinate.)**.** *Let p, q be laws on $\{\pm 1\}^n$ such that $p - q = k\chi_{[n]}u_n$ and the line segment between them consists of laws. Then*

$$\mathcal{Q}_{\boldsymbol{s}}(p\|q) \le \left(\prod_{i=1}^{n} s_i\right) D(p\|q).$$

Corollary 2.2 (The two-bit Heart and total-correlation conjectures follow from the parity contraction.)**.** *Conjecture 2 holds for every two-bit law and arbitrary unequal channel parameters. Consequently Conjecture 1 also holds for every two-bit law and arbitrary unequal channel parameters.*

Theorem 2.3 (Disjoint parity blocks are genuine logical bits whose effective strengths are products of the physical strengths.)**.** *Let $A_1, \dots, A_m \subseteq [n]$ be pairwise disjoint and nonempty, and let*

$$\pi(x) = \big(\chi_{A_1}(x), \dots, \chi_{A_m}(x)\big).$$

Let $\bar{p}, \bar{q}$ be laws on $\{\pm 1\}^m$, and let p, q be their lifts to $\{\pm 1\}^n$, uniform within every fiber of π. Put

$$t_j = \prod_{i \in A_j} s_i.$$

Then

$$\mathcal{C}_{\boldsymbol{s}}(p\|q) = \mathcal{C}_{\boldsymbol{t}}(\bar{p}\|\bar{q}).$$

Moreover there are independent, input-independent flags $Z_1, \dots, Z_m$ and conditional logical trace distances $r_j(Z_j)$ such that

$$\mathcal{Q}_{\boldsymbol{s}}(p\|q) = \mathbb{E}_Z \mathcal{Q}_{\boldsymbol{r}(Z)}(\bar{p}\|\bar{q}), \qquad \mathbb{E} r_j(Z_j) = t_j.$$

The corresponding identities also hold for the trace moments. Consequently the uniform-reference conjectures (Heart and Box) lift from the logical law to the physical parity blocks. More generally, any relative inequality for the pair $(\bar{p}, \bar{q})$ lifts to the lifted pair (p, q); this includes the product-reference conjectures whenever the logical product reference lifts to the physical product reference.

Theorem 2.4 (A nontrivial two-interaction family satisfies the stronger trace-moment conjecture throughout $1 \le \alpha \le 2$.)**.** *Let $A, B \subseteq [n]$ be nonempty and disjoint. For*

$$f = 1 + d\chi_A + a\chi_{A\cup B}, \qquad |a| + |d| \le 1,$$

Conjecture 3 holds for every $1 \le \alpha \le 2$ and arbitrary unequal channel parameters. In four bits this includes every pair–triple, pair–top, and triple–top coordinate plane.

The next two theorems describe the local geometry at the two easiest endpoints. Near the uniform law the full nonlinear problem has a Walsh-diagonal positive Hessian. Near a perfect channel, for an arbitrary fixed law, the first nonzero gap is a positive Dirichlet-type energy on cube edges. These are two views of the same infinitesimal form.

Theorem 2.5 (At the uniform law the full Heart Hessian diagonalizes in Walsh characters with a strict reserve in every nonconstant direction.)**.** *Fix $n \ge 1$ and arbitrary interior channel parameters $0 < s_i < 1$. Put*

$$\mu_i = \sqrt{1 - s_i^2}, \qquad \alpha_i = \operatorname{artanh} \mu_i, \qquad s_U = \prod_{i \in U} s_i,$$

and for nonempty $U \subseteq [n]$ define

$$\vartheta_U := \frac{s_U}{2^{|U|}} \sum_{\varepsilon \in \{\pm 1\}^U} \varphi\left(\sum_{i \in U} \varepsilon_i \alpha_i\right), \qquad \varphi(r) = \frac{r}{\sinh r}, \quad \varphi(0) = 1.$$

If $f_t = 1 + th$ with $\mathbb{E}h = 0$, then

$$\left.\frac{d^2}{dt^2}\right|_0 \mathcal{C}_{\boldsymbol{s}}(f_t) = \sum_{U\neq\varnothing} s_U \widehat{h}(U)^2,$$

$$\left.\frac{d^2}{dt^2}\right|_0 \mathcal{Q}_{\boldsymbol{s}}(f_t) = \sum_{U\neq\varnothing} \vartheta_U \widehat{h}(U)^2.$$

For every nonempty U one has $0 < \vartheta_U < s_U$. Consequently there is an $\varepsilon_{n,\boldsymbol{s}} > 0$ such that every density f with $\|f-1\|_\infty < \varepsilon_{n,\boldsymbol{s}}$ satisfies Heart. This is the actual unpinched inequality, with no antipodal-evenness assumption.

For $x \in \{\pm1\}^n$, let $x^{(i)}$ be obtained by flipping coordinate i, and put

$$\Theta(t) = 1 - \frac{\operatorname{artanh} t}{t} - \frac{1}{2}\log(1-t^2) = \sum_{k\geq 1} \frac{t^{2k}}{2k(2k+1)},$$

with the continuous values $\Theta(0) = 0$ and $\Theta(\pm1) = 1 - \log 2$.

Theorem 2.6 (Near a perfect channel the Heart gap is a positive edge energy on the full Boolean cube, for every input law.)**.** *Let p be an arbitrary law on $\{\pm1\}^n$, write $s_i = \sqrt{1-\mu_i^2}$, and define*

$$\Lambda_i(p) := \frac{1}{2} \sum_{\substack{x\in\{\pm1\}^n \\ x_i=1}} \big(p_x + p_{x^{(i)}}\big)\Theta\left(\frac{p_x - p_{x^{(i)}}}{p_x + p_{x^{(i)}}}\right),$$

where an edge with two zero masses contributes zero. Then, as $\boldsymbol{\mu} \to 0$,

$$\mathcal{G}_{\boldsymbol{s}}(p) = \sum_{i=1}^n \mu_i^2 \Lambda_i(p) + O_p\left(\left(\sum_i \mu_i^2\right)^2\right).$$

In the equal-channel case,

$$\mathcal{G}_{\sqrt{1-\mu^2}}(p) = \mu^2 \Lambda_{Q_n}(p) + O_p(\mu^4), \qquad \Lambda_{Q_n}(p) := \sum_i \Lambda_i(p).$$

The coefficient is nonnegative, it vanishes only at $p = u_n$, and it obeys the coercive estimate

$$\Lambda_{Q_n}(p) \geq \frac{1}{6}\sum_x \big(p_x - 2^{-n}\big)^2.$$

Hence every fixed law satisfies Heart for all equal channel parameters sufficiently close to 1; the inequality is strict there unless the law is uniform.

Corollary 2.7 (Products of disjoint parity-code factors, including boundary codes, have full neighborhoods on which Heart holds.)**.** *Fix pairwise disjoint nonempty blocks $A_1, \dots, A_m$ and interior physical parameters $0 < s_i < 1$. Let*

$$\mathcal{M} = \left\{ \prod_{j=1}^m \big(1 + a_j \chi_{A_j}\big) : -1 \leq a_j \leq 1 \right\}.$$

There is a $\delta > 0$ such that every density f with $\operatorname{dist}_\infty(f, \mathcal{M}) < \delta$ satisfies Heart. In particular this gives full relative neighborhoods of the parity-code densities $1 \pm \chi_A$ and of codes defined by disjoint parity checks.

The third group reorganizes the apparently troublesome odd part. Tensor-power pinching shows that odd information and cross-number-sector coherence are not extensive obstructions. The phase theorem then identifies exactly what odd information means: it is a syndrome of a random phase subset, and the residual difficulty is its correlation with a hidden activation set.

Theorem 2.8 (After amplification, unequal channels reduce to a number-pinched problem with separate evenness in each repeated channel type.)**.** *Fix a base parameter vector $\boldsymbol{s} = (s_1, \dots, s_n)$ and replicate the entire problem N times. Index Walsh subsets by $A = (A_1, \dots, A_n)$ with $A_i \subseteq [N]$, and pinch by the vector of occupation numbers $\boldsymbol{k}(A) = (|A_1|, \dots, |A_n|)$. Let $G_N \cong (\mathbb{Z}_2)^n$ flip all N replicas of each type i simultaneously.*

If, for every base pair $(n, \boldsymbol{s})$, there are errors $\varepsilon_N = o(N)$ such that the vector-number-pinched inequality

$$\mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(F) \le \mathcal{C}_{\boldsymbol{s}^{(N)}}(F) + \varepsilon_N$$

holds for every G_N-invariant density F, then Heart holds for every density on the base n coordinates with parameters $\boldsymbol{s}$. Conversely, full Heart in every dimension implies these pinched inequalities with $\varepsilon_N = 0$. Thus global Heart is equivalent to this classwise-even pinched problem, and sublinear losses are harmless.

Theorem 2.9 (Averaging over any translation subgroup removes exactly a phase syndrome, and the unpaid term is its information about the hidden activation set.)**.** *Identify subsets with $\mathbb{F}_2^n$. Let $H \le \mathbb{F}_2^n$ be a translation subgroup and*

$$p^H(x) = \frac{1}{|H|} \sum_{h \in H} p(x + h).$$

In the random phase experiment associated with the product pure-state channel, let A be the random phase subset, let E be the phase-modulated square-root state, and let

$$Y = A + H^{\perp} \in \mathbb{F}_2^n / H^{\perp}$$

be the phase syndrome. Then

$$D(\rho_p \| \rho_{p^H}) = H(Y \mid E).$$

There is a latent activation set R, with coordinate i active independently with probability $1 - \mu_i$, such that, writing H_R for the projection of H onto R,

$$D(\rho_p \| \rho_{p^H}) = \mathbb{E}_R D\big(p_R \big\| (p_R)^{H_R}\big) + I(Y; R \mid E).$$

The original global-evenization identity is the rank-one case $H = \operatorname{span}\{(1, \dots, 1)\}$, where $Y = |A| \bmod 2$.

Theorem 2.10 (Along an odd perturbation of an even base, every even quantum Taylor coefficient is bounded by full classical reveal.)**.** *For any finite pure-state classical–quantum channel whose involution is implemented by a two-sector output symmetry, take an involution-invariant base law and an involution-odd tangent. In the resulting even Taylor expansion of output relative entropy, every coefficient is at most the corresponding coefficient for fully revealing the classical input.*

Finally, the general mechanisms above support two substantial finite-dimensional victories. The unrestricted four-bit theorem is not claimed: the four-bit result is exactly the antipodally even case.

Theorem 2.11 (Heart holds for every three-bit law with uniform marginals and equal channels.)**.** *Conjecture 1 holds for every three-bit law with uniform one-bit marginals and every equal channel parameter $s \in [0, 1]$.*

Theorem 2.12 (Heart holds for every antipodally even four-bit law and every equal channel.)**.** *Conjecture 1 holds for every antipodally even four-bit law and every equal channel parameter $s \in [0, 1]$.*

3 Common representations and analytic tools

3.1 The Walsh-basis form of the channel

Put

$$a_i = \frac{1 + \mu_i}{2}, \qquad b_i = \frac{1 - \mu_i}{2},$$

and for $A \subseteq [n]$ set

$$\lambda_A = \prod_{i \notin A} a_i \prod_{i \in A} b_i, \qquad \Lambda_{\boldsymbol{s}} = \operatorname{diag}(\lambda_A)_{A \subseteq [n]}.$$

The canonical signal vector has the Walsh expansion

$$|\psi_x\rangle = \sum_{A \subseteq [n]} \sqrt{\lambda_A}\, \chi_A(x) |A\rangle. \tag{1}$$

Indeed, this is simply the tensor-product expansion of the one-coordinate signal states.

For a density f on $\{\pm 1\}^n$, define

$$(C_f)_{A,B} = \widehat{f}(A \triangle B).$$

Then

$$\rho_f := \Phi_{\boldsymbol{s}}(f u_n) = \Lambda_{\boldsymbol{s}}^{1/2} C_f \Lambda_{\boldsymbol{s}}^{1/2}. \tag{2}$$

To verify this, use (1) and compute

$$\langle A|\rho_f|B\rangle = \sqrt{\lambda_A \lambda_B}\, \mathbb{E}[f(X)\chi_A(X)\chi_B(X)] = \sqrt{\lambda_A \lambda_B}\, \widehat{f}(A \triangle B).$$

In particular, the uniform-input output is $\Lambda_{\boldsymbol{s}}$.

It is often useful to write

$$C_f = \sum_{U \subseteq [n]} \widehat{f}(U) X_U,$$

where $X_U|A\rangle = |A \triangle U\rangle$ is translation by U on the subset index.

3.2 Pythagorean identities

Let $\mathcal{E}$ be a trace-preserving conditional expectation and let σ lie in its fixed algebra. Then

$$D(\rho\|\sigma) = D(\rho\|\mathcal{E}\rho) + D(\mathcal{E}\rho\|\sigma). \tag{3}$$

Indeed, $\log(\mathcal{E}\rho)$ and $\log\sigma$ are fixed by $\mathcal{E}$, so their expectations against ρ and $\mathcal{E}\rho$ agree. The same identity holds classically for averaging over a finite symmetry group.

3.3 The BKM Hessian and logarithmic-mean bound

For $C \succ 0$ and Hermitian H, define

$$g_C(H) := \int_0^\infty \mathrm{Tr}\big[H(C+tI)^{-1}H(C+tI)^{-1}\big]\, dt.$$

If $\mathrm{Tr}\, H = 0$, then

$$\left.\frac{d^2}{du^2}\right|_{u=0} D(C+uH\|C) = g_C(H). \tag{4}$$

This follows from the Fréchet derivative

$$D\log_C[H] = \int_0^\infty (C+tI)^{-1} H (C+tI)^{-1}\, dt.$$

In an eigenbasis of C, the BKM kernel is

$$\frac{\log x - \log y}{x-y}.$$

The logarithmic mean is at least the geometric mean, so

$$\frac{\log x - \log y}{x-y} \le \frac{1}{\sqrt{xy}}. \tag{5}$$

For nonnegative scalars define

$$J(a,b) := \int_0^\infty \frac{ab}{(a+t)(b+t)}\, dt = \begin{cases} \dfrac{ab}{a-b}\log\dfrac{a}{b}, & a \neq b, \\ a, & a = b. \end{cases} \tag{6}$$

The continuous boundary value is $J(a,0) = 0$, and J is homogeneous of degree one.

3.4 Entropy and Mellin log-determinant formulas

For positive semidefinite equal-trace matrices X, Y,

$$\operatorname{Tr}(X \log X) - \operatorname{Tr}(Y \log Y) = -\int_0^\infty \log \frac{\det(X + tI)}{\det(Y + tI)}\, dt. \tag{7}$$

One may verify this first for scalars by differentiating twice; equal dimension and equal trace remove the resulting affine ambiguity, after which summing over eigenvalues gives the matrix identity.

More generally, for $\beta = \alpha - 1 \in (0, 1)$,

$$\operatorname{Tr} X^\alpha - \operatorname{Tr} Y^\alpha = -\alpha \frac{\sin(\pi\beta)}{\pi} \int_0^\infty t^\beta \log \frac{\det(X + tI)}{\det(Y + tI)}\, dt. \tag{8}$$

This follows from the scalar identity

$$x^\alpha - y^\alpha = -\alpha \frac{\sin(\pi\beta)}{\pi} \int_0^\infty t^\beta \log \frac{x + t}{y + t}\, dt$$

and the equal-trace cancellation at infinity.

3.5 A complete flagged-parity decomposition

Lemma 3.1 (A physical block carrying only its parity is an input-independent flag followed by a logical binary pure-state channel.)**.** *Let $A \subseteq [n]$ be nonempty. Feed a uniformly random string on $\{\pm 1\}^A$ conditioned on its parity $Y = \chi_A(X)$ into the product channel on A, and call the resulting state Ω_A^Y. Then*

$$\Omega_A^y = \bigoplus_{\{S, S^c\}} m_S |\phi_{y,S}\rangle\langle\phi_{y,S}|,$$

where the sum runs over unordered complementary pairs in 2^A,

$$m_S = \lambda_S + \lambda_{S^c}, \qquad |\phi_{y,S}\rangle = \frac{\sqrt{\lambda_S}|S\rangle + y\sqrt{\lambda_{S^c}}|S^c\rangle}{\sqrt{m_S}}.$$

The flag weights m_S are independent of y. Within flag S, the two logical signal states have trace distance

$$r_S = \frac{2\sqrt{\lambda_S \lambda_{S^c}}}{\lambda_S + \lambda_{S^c}},$$

and

$$\sum_{\{S, S^c\}} m_S r_S = \prod_{i \in A} s_i. \tag{9}$$

Proof. For $S, T \subseteq A$, the matrix entry of the parity-conditioned state is

$$\langle S|\Omega_A^y|T\rangle = \sqrt{\lambda_S \lambda_T}\, \mathbb{E}[\chi_{S \triangle T}(X) \mid \chi_A(X) = y].$$

The conditional expectation is 1 when $S = T$, is y when $T = S^c$, and is 0 otherwise. Thus the matrix splits into the displayed 2×2 complement-pair blocks. Each block has determinant zero and trace m_S, hence is the stated rank-one flagged state.

The overlap between the normalized states with $y = +1$ and $y = -1$ is

$$\frac{\lambda_S - \lambda_{S^c}}{\lambda_S + \lambda_{S^c}},$$

so their trace distance is r_S. Finally,

$$\sqrt{\lambda_S \lambda_{S^c}} = \prod_{i \in A} \sqrt{a_i b_i} = 2^{-|A|} \prod_{i \in A} s_i,$$

independently of S. There are $2^{|A|-1}$ complement pairs, and hence

$$\sum_{\{S, S^c\}} m_S r_S = \sum_{\{S, S^c\}} 2\sqrt{\lambda_S \lambda_{S^c}} = \prod_{i \in A} s_i.$$

□

4 Parity interactions and logical parity blocks

4.1 Top-Walsh contraction

Proof of Theorem 2.1. Let $e(x) = \chi_{[n]}(x)$ and $u = 2^{-n}$. Write

$$p - q = kue, \qquad r_t = q + tkue, \qquad 0 \le t \le 1.$$

We first assume that every r_t is strictly positive and that every $s_i > 0$; the boundary cases follow by approximation, and if some $s_i = 0$ the two channel outputs are identical by cancellation in that coordinate.

Put

$$C_t = \Phi_{\boldsymbol{s}}(r_t), \qquad H = \frac{d}{dt} C_t = \Phi_{\boldsymbol{s}}(kue).$$

For coordinate i,

$$J_i = \frac{|\psi_+^{(i)}\rangle\langle\psi_+^{(i)}| - |\psi_-^{(i)}\rangle\langle\psi_-^{(i)}|}{s_i}$$

is a Hermitian unitary. With

$$J = \bigotimes_{i=1}^{n} J_i, \qquad z = \prod_{i=1}^{n} s_i,$$

tensoring the one-coordinate differences gives

$$H = kuzJ.$$

The second derivative of $D(C_t\|C_0)$ is the BKM form $g_{C_t}(H)$. By (5),

$$g_{C_t}(H) \le k^2u^2z^2 \operatorname{Tr}(C_t^{-1/2} J C_t^{-1/2} J) \le k^2u^2z^2 \left\| C_t^{-1/2} J C_t^{-1/2} \right\|_1. \tag{4.1}$$

Let Ψ be the square matrix whose columns are the 2^n product signal vectors $|\psi_x\rangle$. Then

$$C_t = \Psi \operatorname{diag}(r_t) \Psi^*, \qquad \Psi \operatorname{diag}(e) \Psi^* = zJ.$$

Thus the generalized eigenvalues of the pair (J, C_t), equivalently the eigenvalues of $C_t^{-1/2} J C_t^{-1/2}$, are

$$\frac{e(x)}{z r_t(x)} \qquad (x \in \{\pm 1\}^n).$$

Consequently

$$\left\| C_t^{-1/2} J C_t^{-1/2} \right\|_1 = z^{-1} \sum_x \frac{1}{r_t(x)}.$$

Substituting into (4.1),

$$g_{C_t}(H) \le zk^2u^2 \sum_x \frac{1}{r_t(x)} = z \frac{d^2}{dt^2} D(r_t\|q).$$

Both $D(C_t\|C_0)$ and $D(r_t\|q)$, together with their first derivatives, vanish at $t = 0$. Integrating the second-derivative inequality twice from 0 to 1 proves

$$D(\Phi_{\boldsymbol{s}}(p)\|\Phi_{\boldsymbol{s}}(q)) \le zD(p\|q).$$

□

4.2 The one-bit inequality and its strictness

Lemma 4.1 (For a nontrivial one-bit input bias, an interior pure-state channel contracts strictly more than its trace distance.)**.** *Let Ψ_r be a binary pure-state channel of trace distance $0 \le r \le 1$, let u_1 be the uniform bit, and let ν_z be the bit law of mean z. Then*

$$D(\Psi_r(\nu_z)\|\Psi_r(u_1)) \le rD(\nu_z\|u_1).$$

If $0 < r < 1$ and $z \ne 0$, the inequality is strict.

Proof. Put $\mu = \sqrt{1-r^2}$ and

$$H(x) = h\left(\frac{1+x}{2}\right), \qquad v = \sqrt{\mu^2 + r^2 z^2},$$

where h is binary entropy in natural units. The output under bias z has Bloch length v, while the uniform-input output has Bloch length μ. Since the diagonal of the output is independent of z,

$$D(\Psi_r(\nu_z)\|\Psi_r(u_1)) = H(\mu) - H(v), \qquad D(\nu_z\|u_1) = \log 2 - H(z).$$

It is therefore enough to show that

$$F_r(z) := rH(z) - H(v)$$

is maximized at $z = 0$. The function is even, and direct differentiation gives

$$(1-z^2)F_r''(z) = 1 - r - \frac{\mu^2}{v^2}\left[1 - \frac{1-v^2}{v}\operatorname{artanh} v\right].$$

Set

$$w = \sqrt{1-v^2} = r\sqrt{1-z^2} \le r.$$

The scalar inequality $w \operatorname{artanh} v \le v$ follows from

$$\frac{d}{dv}\left(\frac{v}{\sqrt{1-v^2}} - \operatorname{artanh} v\right) = \frac{1-\sqrt{1-v^2}}{(1-v^2)^{3/2}} \ge 0.$$

Hence the bracket is at least $1 - w$, and

$$\frac{\mu^2}{v^2}(1-w) - (1-r) = \frac{(1-r)(r-w)}{1+w} \ge 0.$$

Thus $F_r'' \le 0$, and $F_r(z) \le F_r(0)$.

If $0 < r < 1$ and $0 < |z| < 1$, then $w < r$, so the final inequality is strict and $F_r''(z) < 0$. Since $F_r'(0) = 0$, this gives $F_r(z) < F_r(0)$ for every nonzero z, with $|z| = 1$ obtained by continuity. □

4.3 Two bits

Proof of Corollary 2.2. Let U, V be the two input bits and let A, B be their pure-state outputs. The joint law of (U, V) and the product of its marginals have the same one-bit marginals, so their difference is purely top-Walsh. Theorem 2.1 gives

$$I(A:B) = D(\rho_{AB}\|\rho_A \otimes \rho_B) \le s_1 s_2 D(P_{UV}\|P_U \otimes P_V) = s_1 s_2 I(U:V).$$

This is TC-Heart for two bits.

For Heart, let σ_i be the uniform-input output of coordinate i. The exact decomposition

$$D(\rho_{AB}\|\sigma_1 \otimes \sigma_2) = D(\rho_A\|\sigma_1) + D(\rho_B\|\sigma_2) + I(A:B)$$

and theorem 4.1 give

$$D(\rho_{AB}\|\sigma_1 \otimes \sigma_2) \le s_1 D(P_U\|u_1) + s_2 D(P_V\|u_1) + s_1 s_2 I(U:V).$$

The right side is exactly

$$s_1(1-s_2)D(P_U\|u_1) + (1-s_1)s_2 D(P_V\|u_1) + s_1 s_2 D(P_{UV}\|u_2),$$

which is the matched two-coordinate reveal functional. □

4.4 Universal block-parity lifting

Proof of Theorem 2.3. We first prove the classical identity. Conditional on a logical parity value $Y_j = \chi_{A_j}(X)$, every proper marginal of the physical string X_{A_j} is uniform and independent of Y_j. Thus the reveal experiment learns the logical bit Y_j exactly when every coordinate of A_j is revealed. This event has probability

$$t_j = \prod_{i \in A_j} s_i,$$

and these events are independent across the disjoint blocks. Any additionally revealed physical bits are uniform nuisance variables with the same law under p and q. Therefore

$$\mathcal{C}_{\boldsymbol{s}}(p\|q) = \mathcal{C}_{\boldsymbol{t}}(\bar{p}\|\bar{q}). \tag{4.2}$$

For the quantum identity, apply theorem 3.1 separately to each block. Conditional on the complete flag $Z = (Z_1, \ldots, Z_m)$, the logical parity vector passes through a product of binary pure-state channels with trace distances $\boldsymbol{r}(Z) = (r_1(Z_1), \ldots, r_m(Z_m))$. The complete flag distribution is independent of the logical input and is the same under $\bar{p}$ and $\bar{q}$. Relative entropy of direct sums with identical block weights is the weighted average of the blockwise relative entropies, so

$$\mathcal{Q}_{\boldsymbol{s}}(p\|q) = \mathbb{E}_Z \mathcal{Q}_{\boldsymbol{r}(Z)}(\bar{p}\|\bar{q}). \tag{4.3}$$

Equation (9) gives $\mathbb{E} r_j(Z_j) = t_j$.

The same argument applies to sandwiched trace moments. If a common flag has weight w_z in both states, then the contribution of that block to the sandwiched moment is w_z times the conditional moment; hence

$$\mathsf{Q}_{\alpha,\boldsymbol{s}}(p\|q) = \mathbb{E}_Z \mathsf{Q}_{\alpha,\boldsymbol{r}(Z)}(\bar{p}\|\bar{q}). \tag{4.4}$$

The classical reveal moment is multilinear in the reveal probabilities. Since the flags are independent across blocks,

$$\mathbb{E}_Z \mathsf{B}_{\alpha,\boldsymbol{r}(Z)}(\bar{p}\|\bar{q}) = \mathsf{B}_{\alpha,\boldsymbol{t}}(\bar{p}\|\bar{q}). \tag{4.5}$$

Combining these identities with any relative logical inequality for $(\bar{p}, \bar{q})$ proves the general lifting assertion. Taking $\bar{q}$ uniform proves the stated Heart and Box consequences. □

As immediate consequences, Heart is closed under tensor products of solved factors, and any law depending only on two disjoint block parities is solved by Corollary 2.2.

4.5 The nested-Walsh trace-moment family

Lemma 4.2 (A two-bit determinant comparison controls the nested family for unequal channels.)**.** *Let*

$$F_{a,d}(x, y) = 1 + dy + axy, \qquad |a| + |d| \le 1.$$

For arbitrary pure-state trace distances s_1, s_2 and every $1 \le \alpha \le 2$, Conjecture 3 holds for this two-bit family.

Proof. By changing the sign of y if necessary, we may assume $d \ge 0$; the sign of a is irrelevant to the determinant calculation below. The endpoint cases follow by continuity, so take $1 < \alpha < 2$ and put $\beta = \alpha - 1$. Hyperbolically normalize

$$s_j = \operatorname{sech} \tau_j, \quad x = e^{\tau_1/\alpha}, \quad y = e^{\tau_2/\alpha}, \quad X = x + x^{-1}, \quad Y = y + y^{-1}.$$

In a suitable basis the density matrix is

$$A_{a,d} = \begin{pmatrix} 1 & d & 0 & a \\ d & 1 & a & 0 \\ 0 & a & 1 & d \\ a & 0 & d & 1 \end{pmatrix},$$

while the channel weight is diagonal with squared entries $xy, x/y, y/x, (xy)^{-1}$. Put $u = a^2$ and $\delta = 1 - d^2$. After removing a common scalar, the weighted and unweighted resolvent determinants are

$$P_D(u; z) = P_0(z) - L_D(z)u + u^2, \qquad P_I(u; z) = C_0(z) - L_I(z)u + u^2,$$

where

$$\begin{aligned} P_0(z) &= z^4 + XYz^3 + [Y^2 + \delta(X^2 - 2)]z^2 + \delta XYz + \delta^2, \\ L_D(z) &= 2(z^2 + 1 + d^2) + XYz, \\ C_0(z) &= (z^2 + 2z + \delta)^2, \\ L_I(z) &= 2(z^2 + 2z + 1 + d^2). \end{aligned}$$

It is enough to prove the pointwise determinant-ratio comparison

$$\frac{P_D(u; z)}{P_D(0; z)} \ge \frac{P_I(u; z)}{P_I(0; z)}. \tag{4.6}$$

Set

$$\Xi = X^2 - 4, \qquad \Upsilon = Y^2 - 4, \qquad \Omega = XY - 4.$$

After cross multiplication, the numerator of the difference in (4.6) is

$$z(A\Xi + B\Omega + C\Upsilon)$$

with $A, C \geq 0$. If $B \geq 0$ there is nothing to prove. If $B < 0$, use

$$\Xi + \Upsilon - 2\Omega = (X - Y)^2 \geq 0$$

to lower-bound the numerator by

$$z[(A + B/2)\Xi + (C + B/2)\Upsilon].$$

The two remaining coefficients are nonnegative. Explicitly,

$$2C + B = ((z+1)^2 - d^2)(2d^2 + 2d + z^2 + 2z) \geq 0,$$

and, with $e = 1 - d \in [0, 1]$,

$$\begin{aligned} 2A + B = {} & z^4 + 4e(2-e)z^3 + e(12 - 7e)z^2 \\ & + 2e(2-e)^3 z + 2e(1-e)(2-e)^2 \geq 0. \end{aligned}$$

Thus (4.6) holds for every $z > 0$. The Mellin formula (8) integrates this pointwise comparison and proves the trace-moment inequality. The base case $a = 0$ is the corresponding one-bit trace-moment inequality and follows from the same 2×2 determinant argument. □

Proof of Theorem 2.4. Let $U = \chi_A$ and $V = \chi_B$. Then

$$f = 1 + dU + aUV$$

is precisely the two-bit density in theorem 4.2. Apply the flagged-parity decomposition to blocks A and B. In every flag pair, theorem 4.2 applies to the two logical pure-state channels. Averaging the quantum inequalities and using the multilinearity of the classical reveal moment, as in (4.4)–(4.5), proves the theorem. □

5 Local geometry at the uniform and perfect-channel endpoints

5.1 The full unpinched Hessian at the uniform law

Proof of Theorem 2.5. Let $f_t = 1 + th$, where $\mathbb{E}h = 0$. On the classical side, the density of the R-marginal is $1 + t\mathbb{E}[h \mid R]$. Hence

$$\frac{d^2}{dt^2}\bigg|_0 \mathcal{C}_{\boldsymbol{s}}(f_t) = \mathbb{E}_R \mathbb{E}\big[\mathbb{E}(h \mid R)^2\big].$$

Since

$$\mathbb{E}(h \mid R) = \sum_{U \subseteq R} \widehat{h}(U)\chi_U,$$

Parseval and independence of the reveal set give

$$\frac{d^2}{dt^2}\bigg|_0 \mathcal{C}_{\boldsymbol{s}}(f_t) = \sum_{U \neq \varnothing} \Pr[U \subseteq R]\widehat{h}(U)^2 = \sum_{U \neq \varnothing} s_U \widehat{h}(U)^2. \tag{10}$$

For the quantum side, write $\Lambda = \Lambda_{\boldsymbol{s}}$. By (2),

$$\rho_{f_t} = \Lambda + t\Delta_h, \qquad \Delta_h = \sum_{U \neq \varnothing} \widehat{h}(U)\Delta_U, \qquad \Delta_U = \Lambda^{1/2} X_U \Lambda^{1/2}.$$

The BKM formula (4) gives

$$\frac{d^2}{dt^2}\bigg|_0 \mathcal{Q}_{\boldsymbol{s}}(f_t) = \int_0^\infty \mathrm{Tr}[\Delta_h(\Lambda + zI)^{-1}\Delta_h(\Lambda + zI)^{-1}]\, dz.$$

There are no mixed terms. Indeed, Λ and its resolvent are diagonal, and for $U \neq V$ the product of translations $X_U X_V = X_{U \triangle V}$ has zero diagonal. Therefore

$$\left.\frac{d^2}{dt^2}\right|_0 \mathcal{Q}_{\boldsymbol{s}}(f_t) = \sum_{U \neq \varnothing} \vartheta_U \widehat{h}(U)^2, \tag{11}$$

where

$$\vartheta_U = \sum_{A \subseteq [n]} J(\lambda_A, \lambda_{A \triangle U}). \tag{12}$$

We now evaluate this coefficient. The homogeneity of J allows us to sum first over coordinates outside U; their weights add to one. For $i \in U$ write

$$a_i = \frac{s_i}{2} e^{\alpha_i}, \qquad b_i = \frac{s_i}{2} e^{-\alpha_i}, \qquad \alpha_i = \operatorname{artanh} \mu_i.$$

For a sign vector $\varepsilon \in \{\pm 1\}^U$, the two complementary weights on U are

$$c e^{r}, \qquad c e^{-r}, \qquad c = \frac{s_U}{2^{|U|}}, \qquad r = \sum_{i \in U} \varepsilon_i \alpha_i.$$

Since

$$J(c e^{r}, c e^{-r}) = c \frac{r}{\sinh r},$$

we obtain

$$\vartheta_U = \frac{s_U}{2^{|U|}} \sum_{\varepsilon \in \{\pm 1\}^U} \varphi\left(\sum_{i \in U} \varepsilon_i \alpha_i \right). \tag{13}$$

Now $0 < \varphi(r) \leq 1$, with equality only at $r = 0$. If $U \neq \varnothing$, the all-plus sign vector has positive sum, so at least one summand in (13) is strictly below one. Hence

$$0 < \vartheta_U < s_U.$$

This proves the Hessian formulas.

It remains to pass from a positive Hessian to an open neighborhood. Let

$$K = \{h : \mathbb{E}h = 0,\ \|h\|_\infty = 1\}.$$

This is compact. Set

$$\delta_* = \min_{\varnothing \neq U \subseteq [n]} (s_U - \vartheta_U) > 0.$$

For $h \in K$, Parseval gives

$$\sum_{U \neq \varnothing} \widehat{h}(U)^2 = \mathbb{E}h^2.$$

Moreover

$$\mathbb{E}h^2 \geq \frac{1}{2^n - 1}. \tag{14}$$

To see this, choose a point where $|h| = 1$. The remaining $2^n - 1$ values have sum of absolute value one; scalar Cauchy–Schwarz yields (14). Therefore the Hessian of the gap is uniformly positive on K:

$$\left.\frac{d^2}{dt^2}\right|_0 \mathcal{G}_{\boldsymbol{s}}(1 + th) \geq \frac{\delta_*}{2^n - 1}.$$

For $|t| \leq 1/2$ and $h \in K$, every classical marginal density lies in $[1/2, 3/2]$. Also C_{1+th} is unitarily similar to $\operatorname{diag}(1+th)$, so

$$\frac{1}{2}\Lambda \preceq \rho_{1+th} \preceq \frac{3}{2}\Lambda.$$

Because the channel parameters are interior, $\Lambda \succ 0$. The classical and quantum functionals therefore have uniformly bounded third derivatives on the compact set $[-1/2, 1/2] \times K$. Taylor's theorem gives an $\varepsilon_{n,\boldsymbol{s}} > 0$ such that

$$\mathcal{G}_{\boldsymbol{s}}(1 + th) \geq 0 \qquad (h \in K,\ |t| < \varepsilon_{n,\boldsymbol{s}}).$$

Writing any nearby density as $f = 1 + th$ with $t = \|f - 1\|_\infty$ proves the neighborhood statement. □

5.2 The full-cube perfect-channel edge energy

Proof of Theorem 2.6. Let

$$\sigma_{\boldsymbol{\mu}} = \bigotimes_{i=1}^{n} \operatorname{diag}\left(\frac{1+\mu_i}{2}, \frac{1-\mu_i}{2}\right)$$

be the uniform-input output. Form the signal matrix whose xth column is $\sqrt{p_x}|\psi_x\rangle$. Its Gram matrix is

$$(B_{\boldsymbol{\mu}})_{x,y} = \sqrt{p_x p_y} \prod_{i:x_i \neq y_i} \mu_i. \tag{15}$$

The output state and $B_{\boldsymbol{\mu}}$ have the same nonzero spectrum. Furthermore every signal state has the same expectation against $\log \sigma_{\boldsymbol{\mu}}$, namely $\operatorname{Tr}(\sigma_{\boldsymbol{\mu}} \log \sigma_{\boldsymbol{\mu}})$. Consequently

$$\mathcal{Q}_{\boldsymbol{s}}(p) = H(\sigma_{\boldsymbol{\mu}}) - H(B_{\boldsymbol{\mu}}). \tag{16}$$

We expand the two sides through total degree two in $\boldsymbol{\mu}$. Since

$$s_i = 1 - \frac{\mu_i^2}{2} + O(\mu_i^4),$$

the reveal experiment either reveals everything or, to quadratic order, misses exactly one coordinate. Thus

$$\mathcal{C}_{\boldsymbol{s}}(p) = D(p\|u_n) + \frac{1}{2}\sum_{i=1}^{n} \mu_i^2 \big[D(p_{-i}\|u_{n-1}) - D(p\|u_n)\big] + O_p\left(\left(\sum_i \mu_i^2\right)^2\right). \tag{17}$$

For the quantum side,

$$H(\sigma_{\boldsymbol{\mu}}) = n\log 2 - \frac{1}{2}\sum_i \mu_i^2 + O\left(\sum_i \mu_i^4\right). \tag{5.6}$$

At $\boldsymbol{\mu} = 0$, $B_{\mathbf{0}} = \operatorname{diag}(p)$. Its first-order perturbation in direction i is supported precisely on the i-edges:

$$(E_i)_{x,x^{(i)}} = \sqrt{p_x p_{x^{(i)}}}.$$

The first derivative of $X \mapsto \operatorname{Tr}(X \log X)$ vanishes on off-diagonal perturbations. Its Hessian at a diagonal matrix is diagonal in matrix units, and an unordered edge with endpoint masses a, b contributes

$$ab\,\ell(a,b), \qquad \ell(a,b) = \frac{\log a - \log b}{a-b},$$

with the continuous interpretation when $a = b$ and with contribution zero if one endpoint mass is zero. Distinct coordinate-edge perturbations have disjoint matrix entries, so there are no quadratic cross-terms. Therefore

$$\mathcal{Q}_{\boldsymbol{s}}(p) = D(p\|u_n) + \sum_{i=1}^{n} \mu_i^2 \sum_{\substack{x\in\{\pm1\}^n \\ x_i=1}} \left[p_x p_{x^{(i)}} \ell(p_x, p_{x^{(i)}}) - \frac{p_x + p_{x^{(i)}}}{2}\right] + O_p\left(\left(\sum_i \mu_i^2\right)^2\right). \tag{18}$$

Here the reference contribution $-1/2$ in (5.6) has been distributed across the i-edges using $\sum_{x_i=1}(p_x + p_{x^{(i)}}) = 1$.

For completeness, the stated fourth-order remainder follows from a symmetry. Changing the sign of a single μ_i conjugates $B_{\boldsymbol{\mu}}$ by the diagonal sign matrix $\operatorname{diag}(x_i)_x$, and therefore leaves its spectrum unchanged. Hence the entropy is even in every μ_i. After the quadratic terms, the next possible total degree is four. If p has zero atoms, the zero rows and columns of (15) may be deleted; on the remaining support the same analytic argument applies.

It remains to identify the edgewise gap. Fix an i-fiber with endpoint masses a, b, and put $m = a + b$. Its coefficient on the classical side of (17) is

$$\frac{1}{2}\big[m\log(m/2) - a\log a - b\log b\big].$$

Its quantum coefficient is

$$ab\ell(a,b) - \frac{m}{2}.$$

Writing

$$a = \frac{m}{2}(1+t), \qquad b = \frac{m}{2}(1-t),$$

a direct simplification gives

$$\frac{1}{2}\big[m\log(m/2) - a\log a - b\log b\big] - ab\ell(a,b) + \frac{m}{2} = \frac{m}{2}\Theta(t). \tag{19}$$

Summing (19) over all coordinate edges proves the anisotropic expansion.

The series

$$\Theta(t) = \sum_{k\geq 1} \frac{t^{2k}}{2k(2k+1)}$$

shows that every edge contribution is nonnegative and that it vanishes exactly when the two endpoint masses agree. In the equal-channel case the cube is connected, so the complete coefficient vanishes only for the uniform law.

Finally, $\Theta(t) \geq t^2/6$, and hence

$$\Lambda_{Q_n}(p) \geq \frac{1}{12} \sum_{\{x,y\}\in E(Q_n)} \frac{(p_x - p_y)^2}{p_x + p_y} \geq \frac{1}{12} \sum_{\{x,y\}\in E(Q_n)} (p_x - p_y)^2.$$

The unnormalized cube Laplacian has spectral gap 2, so

$$\sum_{\{x,y\}\in E(Q_n)} (p_x - p_y)^2 \geq 2\sum_x (p_x - 2^{-n})^2.$$

This proves the coercive estimate and the pointwise neighborhood of the perfect channel. □

Corollary 5.1 (The edge coefficient is the average one-bit reserve of each coordinate conditioned on all the others.)**.** *For*

$$b_i(z) = \mathbb{E}[X_i \mid X_{-i} = z],$$

one has

$$\Lambda_i(p) = \frac{1}{2}\mathbb{E}_{X_{-i}}\Theta\big(b_i(X_{-i})\big).$$

Proof. For the fiber $X_{-i} = z$, let its masses be a, b and $m = a + b = p_{-i}(z)$. Then $b_i(z) = (a-b)/m$, and the contribution of the fiber to Λ_i is $\frac{1}{2}m\Theta(b_i(z))$. Summing over z gives the formula. □

Proposition 5.2 (The uniform-law and perfect-channel expansions meet in the Boolean Dirichlet form.)**.** *Let*

$$p_t(x) = 2^{-n}(1 + th(x)), \qquad \mathbb{E}h = 0.$$

Then

$$\Lambda_{Q_n}(p_t) = \frac{t^2}{12} \sum_{\varnothing\neq U\subseteq[n]} |U|\widehat{h}(U)^2 + O_h(t^3).$$

Equivalently, for equal channels and $|U| = k$,

$$s^k - \vartheta_U = \frac{k}{6}\mu^2 + O(\mu^4) \qquad (s = \sqrt{1-\mu^2}).$$

Proof. On an edge $\{x, y\}$,

$$p_t(x) + p_t(y) = 2^{1-n} + O(t), \qquad \frac{p_t(x) - p_t(y)}{p_t(x) + p_t(y)} = \frac{t}{2}(h(x) - h(y)) + O(t^2).$$

Since $\Theta(r) = r^2/6 + O(r^4)$, the edge contribution is

$$\frac{2^{-n}t^2}{24}(h(x) - h(y))^2 + O_h(t^3).$$

The Fourier identity

$$\sum_{\{x,y\}\in E(Q_n)} (h(x) - h(y))^2 = 2^{n+1}\sum_U |U|\widehat{h}(U)^2$$

gives the first formula. For the second, use

$$\varphi(r) = 1 - \frac{r^2}{6} + O(r^4), \qquad \alpha = \operatorname{artanh}\mu = \mu + O(\mu^3),$$

in (13); the average of $(\sum_{i\in U}\varepsilon_i)^2$ over signs is k. □

5.3 A common neighborhood of the uniform perfect-channel corner

Corollary 5.3 (One neighborhood of the uniform law works uniformly for every sufficiently good equal channel.)**.** *For every fixed n there are $\varepsilon_n > 0$ and $\mu_n > 0$ such that*

$$\|f - 1\|_\infty \le \varepsilon_n, \qquad 0 \le \mu \le \mu_n, \qquad s = \sqrt{1-\mu^2}$$

imply $\mathcal{G}_s(f) \ge 0$.

Proof. Let

$$\Gamma(\mu, t, h) = \mathcal{G}_{\sqrt{1-\mu^2}}(1 + th),$$

where $\mathbb{E}h = 0$ and $\|h\|_\infty = 1$. For $|t| \le 1/2$ and $|\mu|$ small, all relevant classical densities and quantum reference states are uniformly positive, so Γ is smooth.

At $\mu = 0$ the channel is perfect, hence $\Gamma(0, t, h) = 0$. At $t = 0$ both sides vanish and the first t-derivative vanishes. Moreover Γ is even in μ: changing μ to $-\mu$ conjugates the channel output by a tensor product of bit swaps, while s is unchanged. Repeated use of the elementary Hadamard factorization lemma therefore gives

$$\Gamma(\mu, t, h) = \mu^2 t^2 R(\mu, t, h)$$

for a continuous function R.

By theorem 5.2,

$$R(0, 0, h) = \frac{1}{12} \sum_{U \ne \varnothing} |U| \widehat{h}(U)^2 \ge \frac{1}{12} \mathbb{E}h^2 \ge \frac{1}{12(2^n - 1)}.$$

The final inequality is (14). The set of such h is compact, so continuity gives a common rectangle $|\mu| \le \mu_n$, $|t| \le \varepsilon_n$ on which $R \ge 0$. □

5.4 Neighborhoods of products of parity codes

Proof of Corollary 2.7. First consider one block A and the parity-biased density

$$f_a = 1 + a\chi_A, \qquad -1 \le a \le 1.$$

Let ν_a be the logical bit law of mean a and put $t_A = \prod_{i \in A} s_i$. The classical reveal experiment sees the parity exactly when the entire block is revealed, so

$$\mathcal{C}_{\boldsymbol{s}}(f_a) = t_A D(\nu_a \| u_1).$$

By the flagged-parity decomposition,

$$\mathcal{Q}_{\boldsymbol{s}}(f_a) = \mathbb{E}_Z D(\Psi_{r(Z)}(\nu_a) \| \Psi_{r(Z)}(u_1)), \qquad \mathbb{E}r(Z) = t_A.$$

Every flag has positive weight and $0 < r(Z) \le 1$. Because all physical parameters are interior, at least one flag has $r(Z) < 1$. Thus theorem 4.1 gives

$$\mathcal{Q}_{\boldsymbol{s}}(f_a) < t_A D(\nu_a \| u_1) = \mathcal{C}_{\boldsymbol{s}}(f_a) \qquad (a \ne 0). \tag{20}$$

For disjoint blocks, every density in $\mathcal{M}$ is a tensor product of these parity-biased block laws, together with a uniform factor on unused coordinates. Both sides of Heart add under tensor products. Hence the gap is strictly positive at every point of $\mathcal{M}$ except the all-zero parameter point, which is the uniform law. The uniform law has an open Heart neighborhood by Theorem 2.5; every other point has one by continuity and (20). These relative open sets cover the compact manifold $\mathcal{M}$. The distance from $\mathcal{M}$ to the closed complement of their union is positive, and any smaller positive number is a valid tubular radius δ. □

6 Global reductions: number sectors, phase syndromes, and odd perturbations

6.1 Replicated-type number-sector reduction

Proof of Theorem 2.8. Fix a base dimension n, a parameter vector $\boldsymbol{s}$, and an amplification factor N. Index the amplified coordinates by pairs $(i,a) \in [n] \times [N]$, with all replicas of type i carrying parameter s_i. A Walsh basis vector is indexed by

$$A = (A_1, \dots, A_n), \qquad A_i \subseteq [N].$$

Put

$$k_i(A) = |A_i|, \qquad \boldsymbol{k}(A) = (k_1(A), \dots, k_n(A)).$$

Let $P_{\boldsymbol{k}}$ project onto the span of Walsh vectors with occupation vector $\boldsymbol{k}$, and define the vector-number pinching

$$\mathcal{P}_N(X) = \sum_{\boldsymbol{k}} P_{\boldsymbol{k}} X P_{\boldsymbol{k}}.$$

There are $(N+1)^n$ sectors.

Write

$$a_i = \frac{1+\mu_i}{2}, \qquad b_i = \frac{1-\mu_i}{2}.$$

The uniform reference eigenvalue at A is

$$\lambda_A = \prod_{i=1}^{n} a_i^{N-|A_i|} b_i^{|A_i|},$$

which depends only on $\boldsymbol{k}(A)$. Thus the reference is scalar on every vector-number sector.

If A and B lie in the same sector, then

$$|A_i \triangle B_i| \equiv 0 \pmod 2 \qquad \text{for every } i.$$

Let $G_N \cong (\mathbb{Z}_2)^n$ act on amplified inputs by flipping all N replicas of type i simultaneously. Averaging a density F over this group kills exactly those Walsh coefficients for which $|U_i|$ is odd for at least one type i. Since the pinched matrix uses only differences $A \triangle B$ that are even in every type,

$$\mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(F) = \mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(\mathbb{E}_{G_N} F). \tag{21}$$

The classical reveal functional is convex and invariant under the same flips, so

$$\mathcal{C}_{\boldsymbol{s}^{(N)}}(\mathbb{E}_{G_N} F) \le \mathcal{C}_{\boldsymbol{s}^{(N)}}(F). \tag{22}$$

We next control the cost of pinching. By the Pythagorean identity,

$$\mathcal{Q}_{\boldsymbol{s}^{(N)}}(F) = D(\rho_F \| \mathcal{P}_N \rho_F) + \mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(F).$$

A pinching with M blocks satisfies $\rho \le M\mathcal{P}(\rho)$; operator monotonicity of the logarithm therefore gives $D(\rho\|\mathcal{P}\rho) \le \log M$. Here $M = (N+1)^n$, so

$$\mathcal{Q}_{\boldsymbol{s}^{(N)}}(F) \le \mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(F) + n\log(N+1). \tag{23}$$

Now take $F = f^{\otimes N}$, with the tensor factors arranged by replica. Both original functionals are additive:

$$\mathcal{Q}_{\boldsymbol{s}^{(N)}}(f^{\otimes N}) = N\mathcal{Q}_{\boldsymbol{s}}(f), \qquad \mathcal{C}_{\boldsymbol{s}^{(N)}}(f^{\otimes N}) = N\mathcal{C}_{\boldsymbol{s}}(f).$$

Assume the classwise-even pinched inequality with error $\varepsilon_N = o(N)$. Combining (21)–(23),

$$\begin{aligned} N\mathcal{Q}_{\boldsymbol{s}}(f) &\le \mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(f^{\otimes N}) + n\log(N+1) \\ &= \mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(\mathbb{E}_{G_N} f^{\otimes N}) + n\log(N+1) \\ &\le \mathcal{C}_{\boldsymbol{s}^{(N)}}(\mathbb{E}_{G_N} f^{\otimes N}) + \varepsilon_N + n\log(N+1) \\ &\le N\mathcal{C}_{\boldsymbol{s}}(f) + \varepsilon_N + n\log(N+1). \end{aligned}$$

Divide by N and let $N \to \infty$.

Conversely, if Heart holds in every dimension and for arbitrary parameter vectors, then data processing under $\mathcal{P}_N$ gives

$$\mathcal{Q}^{\#}_{\boldsymbol{s}^{(N)}}(F) \le \mathcal{Q}_{\boldsymbol{s}^{(N)}}(F) \le \mathcal{C}_{\boldsymbol{s}^{(N)}}(F)$$

for every F, in particular for every G_N-invariant F. This proves the equivalence. □

For equal channel parameters, one may pinch only by the total number $|A|$. The theorem above shows that equality of the physical parameters is not essential: it merely collapses a vector of occupation numbers to one scalar occupation number.

6.2 The phase experiment

For $A \subseteq [n]$, let

$$w_A = \lambda_A = \prod_{i \notin A} \frac{1+\mu_i}{2} \prod_{i \in A} \frac{1-\mu_i}{2}.$$

For a law p, put

$$|\sqrt{p}\rangle = \sum_x \sqrt{p_x}|x\rangle, \qquad Z_A|x\rangle = \chi_A(x)|x\rangle.$$

The phase experiment samples A with probability w_A and gives the observer the pure state $Z_A|\sqrt{p}\rangle$ in register E.

Lemma 6.1 (Quantum divergence from the uniform output is the uncertainty of the random phase subset.)**.** *In the phase experiment,*

$$\mathcal{Q}_{\boldsymbol{s}}(p) = H(A \mid E).$$

The weighted Gram matrix of the phase ensemble is

$$G^p_{A,B} = \sqrt{w_A w_B}\, \widehat{p}(A \triangle B), \tag{24}$$

and is exactly the Walsh-basis matrix of the channel output.

Proof. Equation (24) follows from

$$\langle\sqrt{p}|Z_A Z_B|\sqrt{p}\rangle = \sum_x p_x \chi_{A\triangle B}(x) = \widehat{p}(A \triangle B).$$

By (2), this weighted Gram matrix is the channel output in the Walsh basis. Hence the average phase state and the channel output have the same nonzero spectrum. Since the conditional phase states are pure,

$$H(A \mid E) = H(w) - H(E).$$

But $w = (w_A)$ is the spectrum of the uniform-input output, so $H(w) = H(\Phi_{\boldsymbol{s}}(u_n))$, while $H(E) = H(\Phi_{\boldsymbol{s}}(p))$. The cross-entropy against the uniform output is independent of the input law, and therefore

$$\mathcal{Q}_{\boldsymbol{s}}(p) = H(\Phi_{\boldsymbol{s}}(u_n)) - H(\Phi_{\boldsymbol{s}}(p)) = H(A \mid E).$$

□

6.3 The phase-syndrome identity

Proof of Theorem 2.9. Averaging p over H retains exactly the Fourier coefficients in the annihilator $H^\perp$:

$$\widehat{p^H}(S) = \begin{cases} \widehat{p}(S), & S \in H^\perp, \\ 0, & S \notin H^\perp. \end{cases} \tag{25}$$

The channel is covariant under input translations, so ρ_{p^H} is the output twirl of ρ_p. Since the uniform reference is fixed by this twirl, the Pythagorean identity gives

$$D(\rho_p \| \rho_{p^H}) = \mathcal{Q}_{\boldsymbol{s}}(p) - \mathcal{Q}_{\boldsymbol{s}}(p^H). \tag{26}$$

By theorem 6.1, the right side is

$$H(A \mid E)_p - H(A \mid E)_{p^H}.$$

Let $Y = A + H^\perp$. If A and B have the same syndrome, then $A \triangle B \in H^\perp$. Equations (24) and (25) show that the conditional phase ensembles given Y have exactly the same Gram matrices under p and p^H. Hence

$$H(A \mid E, Y)_p = H(A \mid E, Y)_{p^H}.$$

For p^H, phase vectors with different syndromes are orthogonal, because every cross-syndrome Gram entry uses a Fourier coefficient outside $H^\perp$. Thus

$$H(Y \mid E)_{p^H} = 0.$$

Using the chain rule $H(A \mid E) = H(Y \mid E) + H(A \mid E, Y)$ in (26) proves

$$D(\rho_p \| \rho_{p^H}) = H(Y \mid E)_p.$$

For the latent-set refinement, put

$$r_i = 1 - \mu_i.$$

Generate R by including coordinate i independently with probability r_i, and conditional on R choose A uniformly among the subsets of R. The marginal inclusion probability of i in A is $r_i/2 = (1-\mu_i)/2$, so this reproduces the phase law w.

Fix R. The phase label is uniform on the group $\mathbb{F}_2^R$, and its Gram matrix is the Fourier-circulant matrix

$$2^{-|R|} \widehat{p}_R(A \triangle B).$$

Fourier transformation diagonalizes this matrix with spectrum p_R. The syndrome map restricted to $\mathbb{F}_2^R$ has kernel

$$H_R^\perp = \{S \subseteq R : \langle S, h_R \rangle = 0 \text{ for all } h_R \in H_R\}.$$

Pinching the phase labels by their syndrome deletes precisely the Fourier frequencies outside $H_R^\perp$, and therefore replaces p_R by its average $(p_R)^{H_R}$. The corresponding entropy gain is

$$H((p_R)^{H_R}) - H(p_R) = D\big(p_R \big\| (p_R)^{H_R}\big),$$

where the final equality is the classical Pythagorean identity for subgroup averaging. This entropy gain is exactly $H(Y \mid E, R)$. Averaging and using

$$H(Y \mid E) = H(Y \mid E, R) + I(Y; R \mid E)$$

gives the refined formula. □

6.4 Coefficientwise full-reveal contraction

Let Ω be a finite label set with an involution $x \mapsto \bar{x}$. Fix probabilities and scores satisfying

$$p_{\bar{x}} = p_x, \qquad z_{\bar{x}} = -z_x.$$

Let $x \mapsto |\psi_x\rangle$ be a pure-state cq channel and put

$$We_x = \sqrt{p_x}|\psi_x\rangle, \qquad Z = \operatorname{diag}(z_x), \qquad \rho = WW^*, \qquad \dot{\rho} = WZW^*.$$

Assume the output space splits into two involution sectors in such a way that ρ is block diagonal and $\dot{\rho}$ is block off diagonal. This is the situation for an even base and an odd tangent. By restricting to $\operatorname{supp}\rho$ and, if necessary, first regularizing and then passing to a limit, it is enough to carry out the analytic expansion with $\rho \succ 0$.

For sufficiently small t, the full-reveal classical law is $p_t(x) = p_x(1 + tz_x)$. Since all odd moments vanish,

$$D(p_t \| p) = \sum_{m \geq 1} \frac{t^{2m}}{m(2m-1)} F_{m-1}, \qquad F_{m-1} = \frac{1}{2} \sum_x p_x z_x^{2m}. \tag{27}$$

On the quantum side, put $R_u = (\rho + uI)^{-1}$. Differentiating relative entropy and using the resolvent formula for the logarithm gives

$$\frac{d}{dt} D(\rho + t\dot{\rho} \| \rho) = \sum_{k \geq 1} (-1)^{k+1} t^k \int_0^\infty \operatorname{Tr}(\dot{\rho} R_u)^{k+1} \, du.$$

Because R_u is block diagonal and $\dot\rho$ is block off diagonal, the trace vanishes when $k+1$ is odd. Therefore

$$D(\rho + t\dot\rho\|\rho) = \sum_{m\geq 1} \frac{t^{2m}}{2m} \int_0^\infty \operatorname{Tr} Y_u^{2m}\, du, \qquad Y_u = R_u^{1/2}\dot\rho R_u^{1/2}.$$

Writing

$$Y_u = \begin{pmatrix} 0 & K_u \\ K_u^* & 0 \end{pmatrix},$$

we have $\operatorname{Tr} Y_u^{2m} = 2\operatorname{Tr}(K_u K_u^*)^m$. Thus

$$D(\rho + t\dot\rho\|\rho) = \sum_{m\geq 1} \frac{t^{2m}}{m(2m-1)} Q_{m-1}, \qquad Q_{m-1} = (2m-1)\int_0^\infty \operatorname{Tr}(K_u K_u^*)^m\, du. \tag{28}$$

We need one matrix inequality.

Lemma 6.2 (A Schatten–ALT inequality controls every even resolvent moment.)**.** *For every linear map V, every real diagonal matrix Z, and every integer $m \geq 1$,*

$$\operatorname{Tr}(V^*ZV)^{2m} \leq \operatorname{Tr}[(VV^*)^{2m}|Z|^{2m}].$$

Proof. Write $Z = U|Z|$, where U is a self-adjoint unitary commuting with $|Z|$, and set $C = |Z|^{1/2}V$. Then

$$V^*ZV = C^*UC.$$

Schatten Hölder gives

$$\|C^*UC\|_{2m} \leq \|C^*\|_{4m}\,\|UC\|_{4m} = \|C\|_{4m}^2.$$

Raising to the power $2m$,

$$\operatorname{Tr}|C^*UC|^{2m} \leq \operatorname{Tr}(CC^*)^{2m}.$$

Now

$$CC^* = |Z|^{1/2}VV^*|Z|^{1/2}.$$

The Araki–Lieb–Thirring inequality with exponent $2m$ yields

$$\operatorname{Tr}(|Z|^{1/2}VV^*|Z|^{1/2})^{2m} \leq \operatorname{Tr}(VV^*)^{2m}|Z|^{2m}.$$

Since V^*ZV is self-adjoint, its even power equals its absolute even power. □

Proof of Theorem 2.10. For $u > 0$, set

$$V_u = W^*(\rho + uI)^{-1/2}, \qquad \widetilde{Y}_u = V_u^* Z V_u = Y_u.$$

Also put

$$C = W^*W, \qquad B_u = V_u V_u^*.$$

The push-through identity gives

$$B_u = W^*(WW^* + uI)^{-1}W = C(C+uI)^{-1}.$$

By theorem 6.2,

$$\operatorname{Tr} Y_u^{2m} \leq \operatorname{Tr} B_u^{2m}|Z|^{2m}.$$

Spectral calculus gives the exact identity

$$(2m-1)\int_0^\infty B_u^{2m}\, du = C,$$

because for every scalar $c \geq 0$,

$$(2m-1)\int_0^\infty \left(\frac{c}{c+u}\right)^{2m} du = c.$$

Consequently

$$(2m-1)\int_0^\infty \operatorname{Tr} Y_u^{2m}\, du \leq \operatorname{Tr} C|Z|^{2m} = \sum_x p_x z_x^{2m},$$

since $C_{xx} = p_x$. Using $\operatorname{Tr} Y_u^{2m} = 2\operatorname{Tr}(K_u K_u^*)^m$ and the definition in (28),

$$Q_{m-1} \leq \frac{1}{2}\sum_x p_x z_x^{2m} = F_{m-1}.$$

This holds for every $m \geq 1$, proving the coefficientwise comparison. □

6.5 The isolated one-coordinate removal problem

The phase theorem identifies a precise next target. Fix a coordinate i and average only over flipping that coordinate:

$$q(x) = \frac{1}{2}\big(p(x) + p(x^{(i)})\big) = u_i \otimes p_{-i}.$$

For $S \subseteq [n] \setminus \{i\}$ put

$$d_i(S) := D\big(p_{S\cup\{i\}} \big\| u_i \otimes p_S\big)\,.$$

Proposition 6.3 (One-coordinate removal is equivalent to paying a single hidden-set information term.)**.** *Let A_i be the indicator that the random phase subset contains i, and put $r_j = 1 - \mu_j$. Then*

$$D(\rho_p \| \rho_q) = H(A_i \mid E)$$

and

$$H(A_i \mid E) = r_i \mathbb{E}_{S\sim \mathrm{Ber}(\boldsymbol{r}_{-i})} d_i(S) + I(A_i; R \mid E). \tag{29}$$

The classical random-reveal increment is

$$\mathcal{C}_{\boldsymbol{s}}(p) - \mathcal{C}_{\boldsymbol{s}}(q) = s_i \mathbb{E}_{S\sim \mathrm{Ber}(\boldsymbol{s}_{-i})} d_i(S). \tag{30}$$

Hence the desired one-coordinate removal inequality is reduced exactly to

$$I(A_i; R \mid E) \le s_i \mathbb{E}_{\boldsymbol{s}_{-i}} d_i(S) - r_i \mathbb{E}_{\boldsymbol{r}_{-i}} d_i(S). \tag{31}$$

The right side is nonnegative.

Proof. Apply Theorem 2.9 with $H = \mathrm{span}\{e_i\}$. Its syndrome is exactly A_i, and its subgroup average is q, proving the first identity. Conditional on the latent activation set, the projected subgroup is trivial unless $i \in R$; when $i \in R$, it averages the ith coordinate. The refined phase identity therefore becomes (29).

For the classical increment, if coordinate i is not revealed then p and q have the same revealed marginal. If i is revealed and the other revealed coordinates form S, their relative divergence is $d_i(S)$. This proves (30).

Finally, $d_i(S)$ is monotone under enlarging S, by data processing, and $r_j = 1 - \mu_j \le \sqrt{1-\mu_j^2} = s_j$. A monotone coupling therefore gives

$$r_i \mathbb{E}_{\boldsymbol{r}_{-i}} d_i(S) \le s_i \mathbb{E}_{\boldsymbol{s}_{-i}} d_i(S),$$

so the right side of (31) is nonnegative. Comparing (29) and (30) proves the equivalence. □

Inequality (31) remains open. It is narrower than a general quantum side-information inequality: the unknown is one phase bit, the hidden variable is a Bernoulli activation set, and the available budget is exactly the increase obtained by moving reveal probabilities from $1-\mu_j$ to $\sqrt{1-\mu_j^2}$. The coefficientwise theorem shows that uncontrolled high Taylor coefficients are not the basic obstruction; the difficulty is paying for this structured hidden-set information using only the incremental random-reveal budget.

7 Proof of the three-bit theorem

7.1 Removing the cubic interaction

Let u_n denote the uniform law on $\Omega_n = \{\pm 1\}^n$. Write a law as $p = f u_n$, with $f \ge 0$ and $\mathbb{E} f = 1$. Fix an equal channel parameter $s \in [0,1]$, put

$$\mu = \sqrt{1-s^2}, \qquad |\psi_x\rangle = \sqrt{\frac{1+\mu}{2}}|0\rangle + x\sqrt{\frac{1-\mu}{2}}|1\rangle,$$

and let σ_n be the uniform-input output. The endpoint cases $s = 0, 1$ are equalities. We henceforth assume $0 < s < 1$ and write

$$s = \operatorname{sech}\tau, \qquad \mu = \tanh\tau, \qquad \tau > 0.$$

Uniform one-bit marginals give the Walsh form

$$f_d = 1 + a\chi_{12} + b\chi_{13} + c\chi_{23} + d\chi_{123}. \tag{32}$$

Let f_0 be the same density with $d = 0$, and put $p_d = f_d u_3$, $p_0 = f_0 u_3$. Global complementation changes d to $-d$. Averaging over this symmetry sends p_d to p_0, and channel covariance sends the corresponding output to the quantum twirl. Since the uniform output is fixed, the Pythagorean identity gives

$$D(\Phi_s(p_d)\|\sigma_3) = D(\Phi_s(p_d)\|\Phi_s(p_0)) + D(\Phi_s(p_0)\|\sigma_3). \tag{33}$$

All proper marginals of p_d and p_0 agree, so only full reveal sees their difference. Hence

$$\mathcal{C}_s(p_d) = \mathcal{C}_s(p_0) + s^3 D(p_d\|p_0). \tag{34}$$

Theorem 2.1, applied to the pure χ_{123} difference, gives

$$D(\Phi_s(p_d)\|\Phi_s(p_0)) \le s^3 D(p_d\|p_0).$$

Combining with (33)–(34),

$$\mathcal{G}_s(p_d) \ge \mathcal{G}_s(p_0). \tag{35}$$

Checkpoint. It is enough to prove the theorem for globally even laws, i.e. for $d = 0$.

7.2 Even laws are four masses

When $d = 0$, the law is constant on antipodal pairs. Its four pair-masses are

$$\begin{aligned} q_1 &= \frac{1+a+b+c}{4}, \quad q_2 = \frac{1+a-b-c}{4}, \\ q_3 &= \frac{1-a+b-c}{4}, \quad q_4 = \frac{1-a-b+c}{4}. \end{aligned} \tag{36}$$

Thus admissibility is exactly $q = (q_1, q_2, q_3, q_4) \in \Delta_4$.

The classical side has a particularly symmetric form:

$$\mathcal{C}_s(q) = s^2(1-s)\sum_{i<j}(q_i+q_j)\log\big(2(q_i+q_j)\big) + s^3\sum_i q_i \log(4q_i). \tag{37}$$

Indeed, a two-coordinate reveal induces one of the three binary partitions of the four antipodal classes, and the six pair sums in (37) are precisely the two sides of those three partitions.

For the quantum side, introduce

$$M = \begin{pmatrix} 1 & a & b & c \\ a & 1 & c & b \\ b & c & 1 & a \\ c & b & a & 1 \end{pmatrix}, \qquad H_0 = \operatorname{diag}(3,-1,-1,-1).$$

The two parity blocks are

$$M_\varepsilon = e^{\varepsilon\tau H_0/2} M e^{\varepsilon\tau H_0/2}, \qquad \varepsilon \in \{+1,-1\},$$

and direct block decomposition gives

$$\mathcal{Q}_s(q) = \frac{s^3}{8}\sum_{\varepsilon=\pm1} \operatorname{Tr} M_\varepsilon(\log M_\varepsilon - \varepsilon\tau H_0).$$

Let U be the normalized 4×4 Hadamard matrix and $p = (1,1,1,1)^T/2$. Then

$$U^*MU = 4Q, \qquad U^*H_0U = 4pp^T - I, \qquad Q = \operatorname{diag}(q_1,q_2,q_3,q_4).$$

Put $v = (\sqrt{q_1},\dots,\sqrt{q_4})^T$ and

$$\eta_\varepsilon = \frac{e^{4\varepsilon\tau}-1}{4}, \qquad N_\varepsilon = Q + \eta_\varepsilon vv^T.$$

The eigenvalues of M_ε are those of $4e^{-\varepsilon\tau}N_\varepsilon$: use that ABA and $B^{1/2}A^2B^{1/2}$ have the same spectrum. The remaining terms in the relative entropy are affine in q and hence depend only on $\sum_i q_i = 1$. Therefore the q-dependent part of the quantum functional is

$$\frac{s^3}{2}\sum_{\varepsilon=\pm1} e^{-\varepsilon\tau}\operatorname{Tr} N_\varepsilon \log N_\varepsilon. \tag{38}$$

This representation makes the exceptional full permutation symmetry in $q_1, \dots, q_4$ transparent.

Let

$$G(q) = \mathcal{G}_s(q).$$

It is continuous on Δ_4, analytic on the relative interior of each face, symmetric in the four masses, and satisfies $G(1/4, 1/4, 1/4, 1/4) = 0$.

Checkpoint. The theorem has become a minimization problem for one symmetric function on a tetrahedron.

7.3 Compare two masses directly

Fix two distinct positive masses x and y; call the other two masses z and u. For $c \geq 0$, define the shifted logarithmic mean

$$L_c(x, y) = \int_0^\infty \frac{dt}{(t + x + c)(t + y + c)} = \frac{\log(x + c) - \log(y + c)}{x - y}, \tag{39}$$

and the geometric reserve

$$C_{x,y;z,u} = L_z(x, y) + L_u(x, y) - L_0(x, y). \tag{40}$$

For $t > 0$, also put

$$D(t) = \frac{1}{(t + x)(t + y)}, \qquad R(t) = \frac{t(2t + x + y)}{2(t + x)(t + y)}, \tag{41}$$

$$\theta(t) = 1 - \frac{t}{2}\left(\frac{1}{t + x} + \frac{1}{t + y} + \frac{1}{t + z} + \frac{1}{t + u}\right). \tag{42}$$

Finally, use the channel coordinate

$$w = \tanh^2(2\tau), \qquad \kappa(w) = \frac{1 - \sqrt{1 - w}}{w}, \qquad A(w) = \frac{1 - s}{w}, \tag{43}$$

with the continuous values $\kappa(0) = 1/2$ and $A(0) = 1/8$.

7.3.1 The exact pair formula

Proposition 7.1 (Pair divided gradient)**.** *Let $q_i = x$, $q_j = y$. Then*

$$\boxed{\frac{\partial_i G - \partial_j G}{q_i - q_j} = s^2 w \left[A(w) C_{x,y;z,u} + \int_0^\infty D(t) R(t) \frac{\theta(t) + \kappa(w)}{1 - w\theta(t)^2}\, dt\right].} \tag{44}$$

Proof. The classical formula (37) gives

$$\frac{\partial_i \mathcal{C}_s - \partial_j \mathcal{C}_s}{x - y} = s^2(1 - s)(L_z + L_u) + s^3 L_0. \tag{45}$$

For the quantum side, the matrix determinant lemma gives

$$\det(N_\varepsilon + tI) = \prod_k (t + q_k)\left(1 + \eta_\varepsilon \sum_k \frac{q_k}{t + q_k}\right).$$

Since

$$\sum_k \frac{q_k}{t + q_k} = 2(1 + \theta(t)),$$

a direct differentiation in the trace-zero direction $e_i - e_j$ gives

$$-\frac{(\partial_i - \partial_j) \log\det(N_\varepsilon + tI)}{x - y} = D(t)\left[1 + \frac{2\eta_\varepsilon R(t)}{1 + 2\eta_\varepsilon(1 + \theta(t))}\right]. \tag{46}$$

We use the elementary equal-trace identity

$$\operatorname{Tr} X \log X - \operatorname{Tr} Y \log Y = -\int_0^\infty \log \frac{\det(X + tI)}{\det(Y + tI)}\, dt. \tag{47}$$

To verify it, differentiate twice in a scalar eigenvalue: both sides have second derivative $1/x$; equal dimension and trace cancel the resulting affine terms.

The constant term in (46), after summing the two parity branches in (38), is

$$\frac{s^3}{2}(e^{-\tau}+e^{\tau})L_0 = s^2 L_0.$$

Write $r=\tanh(2\tau)=\sqrt{w}$. Since

$$1+2\eta_+(1+\theta)=\frac{1+r\theta}{1-r}, \qquad 1+2\eta_-(1+\theta)=\frac{1-r\theta}{1+r},$$

one obtains

$$\sum_{\varepsilon=\pm 1} e^{-\varepsilon\tau}\frac{\eta_\varepsilon}{1+2\eta_\varepsilon(1+\theta)} = -\frac{w}{s}\frac{\theta+\kappa(w)}{1-w\theta^2}. \tag{48}$$

Subtracting the quantum divided gradient from (45), and using $1-s=wA(w)$, proves (44). □

The formula has only one potentially negative ingredient: the scalar reserve $C_{x,y;z,u}$. The integral can be controlled coefficient by coefficient with three short lemmas.

7.3.2 The channel expansion has one moving root

The Catalan expansion is

$$\kappa(w)=\sum_{j\ge 0} a_j w^j, \qquad a_j=\frac{1}{2^{2j+1}(j+1)}\binom{2j}{j}>0, \qquad \sum_{j\ge 0} a_j = 1. \tag{49}$$

Also

$$A(w)=\sum_{n\ge 0} b_n w^n, \qquad b_n>0. \tag{50}$$

A convenient proof of the second assertion is the following. Let $K(u)=(1-\sqrt{1-u})/u$. The identities

$$\tanh\tau=\sqrt{w}\,K(w), \qquad 1-\operatorname{sech}\tau=\tanh\tau\tanh(\tau/2)$$

give

$$A(w)=K(w)^2K\big(wK(w)^2\big),$$

which is a composition of power series with positive coefficients.

For $n\ge 0$, define

$$Q_n(t)=t^{2n+1}+\sum_{j=0}^{n} a_j t^{2(n-j)}. \tag{51}$$

Lemma 7.2 (One root)**.** *For each n, Q_n has exactly one root $-\rho_n$ in $(-1,0)$, and*

$$\frac{1}{2}=\rho_0<\rho_1<\rho_2<\cdots.$$

Moreover, for some $c_n>0$,

$$Q_n(t)\ge c_n(t+\rho_n) \qquad (-1\le t\le 1). \tag{52}$$

Finally,

$$\rho_1>\frac{2}{3}. \tag{53}$$

Proof. For $r>0$,

$$\frac{Q_n(-r)}{r^{2n}}=-r+\sum_{j=0}^{n} a_j r^{-2j}$$

is strictly decreasing. It is positive near 0 and is negative at $r=1$, because $\sum_{j=0}^{n} a_j<1$. This proves uniqueness. The recurrence

$$Q_{n+1}(t)=t^2Q_n(t)+a_{n+1}$$

shows $Q_{n+1}(-\rho_n) > 0$, hence $\rho_{n+1} > \rho_n$. The quotient $Q_n(t)/(t+\rho_n)$ is continuous and strictly positive on $[-1,1]$, which gives (52). Finally,

$$\rho_1^2\left(\rho_1 - \frac{1}{2}\right) = \frac{1}{8}.$$

At $2/3$ the left side is $2/27 < 1/8$, proving (53). □

7.3.3 A universal weighted-mean estimate

Lemma 7.3 (The score cannot lean too far left)**.** *With the notation of theorem 7.1,*

$$-\frac{\int_0^\infty D(t)R(t)\theta(t)\,dt}{\int_0^\infty D(t)R(t)\,dt} \le \frac{2}{3}. \tag{54}$$

Proof. Put

$$R_{z,u}(t) = \frac{t}{2}\left(\frac{1}{t+z} + \frac{1}{t+u}\right) \le 1.$$

Then $\theta = 1 - R - R_{z,u}$, so $-\theta \le R$. It is therefore enough to prove

$$\frac{\int DR^2}{\int DR} \le \frac{2}{3}. \tag{55}$$

Since $R = -tD'/(2D)$, integration by parts gives

$$\int_0^\infty DR\,dt = \frac{1}{2}\int_0^\infty D\,dt. \tag{56}$$

Write

$$d = \frac{x+y}{2}, \qquad \rho = \frac{x-y}{x+y}, \qquad a = \rho^2, \qquad t = d\frac{v}{1-v}.$$

Then

$$D(t)\,dt = \frac{1}{d}\frac{dv}{1-a(1-v)^2}, \qquad R(t) = \frac{v}{1-a(1-v)^2}.$$

Expanding the denominators and integrating beta monomials gives

$$\int_0^\infty D(t)\,dt = \frac{1}{d}\sum_{m\ge 0}\frac{a^m}{2m+1},$$
$$\int_0^\infty D(t)R(t)^2\,dt = \frac{1}{d}\sum_{m\ge 0}\frac{m+2}{2(2m+1)(2m+3)}a^m.$$

For every $m \ge 0$,

$$\frac{m+2}{2(2m+3)} \le \frac{1}{3}.$$

Thus $\int DR^2 \le \frac{1}{3}\int D$, and (55) follows from (56). □

Combining theorems 7.2 and 7.3 gives, for every $n \ge 1$,

$$\int_0^\infty D(t)R(t)Q_n(\theta(t))\,dt > 0. \tag{57}$$

Indeed, use (52) and $\rho_n \ge \rho_1 > 2/3$.

7.3.4 The constant coefficient is a convex edge energy

Define

$$\mathfrak{c}(a,b)=\sum_{k\ge 1}\frac{(a-b)^{2k}}{2k(2k+1)(a+b)^{2k-1}},\qquad \mathcal{C}(q)=\sum_{i<j}\mathfrak{c}(q_i,q_j). \tag{58}$$

The continuous value at $a=b=0$ is zero.

Lemma 7.4 (Perfect-channel coefficient)**.** *For distinct positive q_i,q_j,*

$$\Gamma_0^{ij}:=\frac{1}{8}C_{x,y;z,u}+\int_0^\infty D(t)R(t)\left(\theta(t)+\frac{1}{2}\right)dt=\frac{1}{8}\frac{\partial_i\mathcal{C}-\partial_j\mathcal{C}}{q_i-q_j}>0. \tag{59}$$

Proof. We first identify the second-order gap at the perfectly distinguishing endpoint. Put

$$\ell(a,b)=\frac{\log a-\log b}{a-b},$$

with the continuous value $1/a$ at $a=b$. Since $s^2(1-s)=\tau^2/2+O(\tau^4)$ and $s^3=1-3\tau^2/2+O(\tau^4)$, the coefficient of τ^2 on the classical side is

$$\frac{1}{2}\sum_{i<j}(q_i+q_j)\log\bigl(2(q_i+q_j)\bigr)-\frac{3}{2}\sum_i q_i\log(4q_i).$$

For the quantum side, use (38). One has

$$N_\varepsilon=Q+\varepsilon\tau vv^T+2\tau^2vv^T+O(\tau^3).$$

The Hessian of $X\mapsto \operatorname{Tr}X\log X$ at the diagonal matrix Q gives

$$g_Q(vv^T)=1+2\sum_{i<j}q_iq_j\ell(q_i,q_j).$$

Expanding the two branches shows that the quantum coefficient is

$$\sum_{i<j}q_iq_j\ell(q_i,q_j)-\frac{3}{2};$$

the constant $-3/2$ is fixed by the uniform law, whose quantum divergence is identically zero.
For two positive numbers a,b, set

$$\gamma(a,b)=\frac{1}{2}\bigl[(a+b)\log(2(a+b))-a\log(4a)-b\log(4b)\bigr]\\ -ab\ell(a,b).$$

Writing $r=(a-b)/(a+b)$ and using the series for $\operatorname{artanh}r$ gives the elementary identity

$$\gamma(a,b)=\frac{1}{2}\mathfrak{c}(a,b)-\frac{1}{2}(a+b).$$

Since $\sum_{i<j}(q_i+q_j)=3$, the linear terms cancel the quantum constant. Therefore

$$G(q,\operatorname{sech}\tau)=\frac{\tau^2}{2}\mathcal{C}(q)+O(\tau^4). \tag{60}$$

Now $w=4\tau^2+O(\tau^4)$ and $s^2=1+O(\tau^2)$. Comparing the constant term of (44) with (60) gives the equality in (59). Each summand of (58) is a positive multiple of

$$(a+b)\left(\frac{a-b}{a+b}\right)^{2k},$$

the perspective of the convex function $r\mapsto r^{2k}$. Thus $\mathcal{C}$ is symmetric and convex. Along the transposition line $(q_i,q_j)=(m+r,m-r)$ it is even and strictly convex; strictness already comes from the $k=1$ term involving the pair (i,j). Hence $(q_i-q_j)(\partial_i\mathcal{C}-\partial_j\mathcal{C})>0$. □

We can now read the sign of the entire pair formula.

Proposition 7.5 (Ordered-pair criterion)**.** *If*

$$C_{x,y;z,u} \geq 0,$$

then

$$\frac{\partial_i G - \partial_j G}{q_i - q_j} > 0 \qquad (0 < s < 1). \tag{61}$$

Proof. Expanding (44) with (49)–(50) gives

$$\frac{\partial_i G - \partial_j G}{(q_i - q_j)s^2 w} = \sum_{n \geq 0} \left[b_n C_{x,y;z,u} + \int_0^\infty DR\, Q_n(\theta)\, dt \right] w^n.$$

The coefficient $n = 0$ is positive by theorem 7.4. Every coefficient $n \geq 1$ is positive by (57) and $b_n C_{x,y;z,u} \geq 0$. □

There are two simple situations in which the geometric reserve has the required sign.

Lemma 7.6 (Two ways to make the reserve nonnegative)**.** *The following implications hold.*

1. *If $x > y \geq z, u \geq 0$, then $C_{x,y;z,u} > 0$.*
2. *If the four masses are (x, x, y, z) with $y \geq z$ and the selected pair is one x and the mass y, then $C_{x,y;x,z} \geq 0$.*

Proof. The function $c \mapsto L_c(x, y)$ decreases. Under the first hypothesis,

$$C_{x,y;z,u} \geq 2L_y - L_0 = \frac{1}{x - y} \log \frac{(x + y)^2}{4xy} > 0.$$

For the second assertion,

$$C_{x,y;x,z} = \frac{1}{x - y} \log \frac{2y(x + z)}{(x + y)(y + z)}.$$

The numerator minus the denominator inside the logarithm is

$$2y(x + z) - (x + y)(y + z) = (x - y)(y - z).$$

Thus the logarithm has the same sign as $x - y$, proving the claim. □

> **Checkpoint.** Whenever the two selected masses are the two largest, or one belongs to a repeated pair in the ordered configuration (x, x, y, z), their divided gradient has the strict sign dictated by their order.

7.4 KKT forces one small mass and three equal masses

Let q^* minimize G on the compact simplex, and relabel its coordinates so that

$$q_1 \geq q_2 \geq q_3 \geq q_4 \geq 0.$$

On the relative interior of the support, the KKT equations say that the partial derivatives corresponding to positive masses are equal.

Suppose first that $q_1 > q_2 > 0$. The selected masses q_1, q_2 are the two largest, so theorems 7.5 and 7.6 give

$$\frac{\partial_1 G - \partial_2 G}{q_1 - q_2} > 0,$$

contradicting KKT. Hence

$$q_1 = q_2 = x. \tag{62}$$

If the support has at least three points and $q_2 > q_3$, write $q = (x, x, y, z)$ with $y = q_3 \geq z = q_4$. Comparing one x with y and using the second part of theorem 7.6 again contradicts KKT. Thus every minimizer with support at least three has

$$q_1 = q_2 = q_3. \tag{63}$$

There remain three boundary orbits besides the line in (63). They are already controlled by the same pair criterion.

Two equal pairs. Let

$$q(r) = \left(\frac{1+r}{4}, \frac{1+r}{4}, \frac{1-r}{4}, \frac{1-r}{4}\right), \qquad 0 \le r \le 1.$$

For a selected large and small mass, the reserve in the second part of theorem 7.6 is zero. Hence

$$G'(r) = \frac{1}{2}(\partial_{q_1} G - \partial_{q_3} G) > 0 \qquad (r > 0).$$

Since $G(0) = 0$, this whole line is nonnegative. Its endpoint is the only possible support-two minimizer.

One large mass. Let

$$q(r) = \left(\frac{1+3r}{4}, \frac{1-r}{4}, \frac{1-r}{4}, \frac{1-r}{4}\right), \qquad 0 \le r \le 1.$$

The large mass and any small mass satisfy the first part of theorem 7.6; therefore

$$G'(r) = \frac{3}{4}(\partial_{q_1} G - \partial_{q_2} G) > 0 \qquad (r > 0).$$

This line is nonnegative, including its support-one endpoint.
Consequently a negative minimum could only lie on

$$q = (p, m, m, m), \qquad p = \frac{1-3\rho}{4}, \qquad m = \frac{1+\rho}{4}, \qquad 0 \le \rho \le \frac{1}{3}. \tag{64}$$

Here one mass is smaller than the other three. This is the only ordering for which the geometric reserve in (40) can be negative.

Checkpoint. The original three-dimensional minimization problem has been reduced to the single interval (64).

7.5 The remaining line is a two-by-two calculation

Put

$$z = \frac{\rho}{1-\rho} \in \left[0, \frac{1}{2}\right], \qquad \eta = e^{-\tau}, \qquad \lambda_\pm = (\cosh(2\tau) \pm z \sinh(2\tau))^{-1/2}. \tag{65}$$

Let

$$h(r) = \frac{(1+r)\log(1+r) + (1-r)\log(1-r)}{2}.$$

For every physical $\lambda > 0$, define

$$u_\lambda = \sqrt{1-(1-z^2)\lambda^4}, \qquad v_\lambda = \sqrt{1-(1-4z^2)\lambda^4}, \tag{66}$$

$$\mathscr{D}_z(\lambda) = \lambda^{-2}\big(h(v_\lambda) - h(u_\lambda)\big), \qquad E(z) = h(2z) - h(z), \tag{67}$$

$$R(z) = \frac{1}{2}\left[8(1+z)\log(1+z) + (1-2z)\log(1-2z) - 3(1+2z)\log(1+2z)\right]. \tag{68}$$

Lemma 7.7 (Exact block identity)**.** *Along* (64)*,*

$$\boxed{\frac{4(1+z)}{s^3} G(q) = (\eta + \eta^{-1})E(z) + (2 - \eta - \eta^{-1})R(z) - \eta^{-1}\mathscr{D}_z(\lambda_+) - \eta\mathscr{D}_z(\lambda_-).} \tag{69}$$

Proof. After changing signs of coordinates if necessary, the even density is $1 - \rho(\chi_{12} + \chi_{13} + \chi_{23})$. In parity branch $\ell = e^{\pm\tau}$, two eigenvalues are $\ell^{-1}(1+\rho)$, while the remaining two form

$$B_\ell = \begin{pmatrix} \ell^3 & -\sqrt{3}\,\ell\rho \\ -\sqrt{3}\,\ell\rho & \ell^{-1}(1-2\rho) \end{pmatrix}.$$

If a positive 2×2 matrix has trace m and eigenvalue asymmetry r, then

$$\operatorname{Tr} B \log B = m\log(m/2) + mh(r). \tag{70}$$

For B_ℓ, the trace is

$$\frac{2\ell}{1+z}\big(\cosh(2\tau) \pm z\sinh(2\tau)\big),$$

and its determinant is

$$\ell^2\frac{(1+2z)(1-2z)}{(1+z)^2}.$$

Thus its asymmetry is $v_{\lambda_\pm}$. Deleting the off-diagonal entries gives the same trace and asymmetry $u_{\lambda_\pm}$. Formula (70) shows that the two mixing increments are exactly the last two terms of (69). Collecting the remaining scalar logarithms gives $E(z)$ and $R(z)$. No matrix larger than 2×2 remains. □

We now prove exactly the scalar information needed to use this identity.

7.5.1 Endpoint reserve, slope, and a chord

Lemma 7.8 (The scalar toolkit)**.** *For $0 \le z \le 1/2$:*

1. $0 \le R(z) \le E(z)$;
2. $\mathscr{D}_z'(1) \ge R(z)$;
3. *for $0 < \lambda \le 1$,*
$$\mathscr{D}_z(\lambda) \le E(z) - (1-\lambda)R(z); \tag{71}$$
4. $\eta^{-1}\lambda_+ + \eta\lambda_- \le 2$;
5. *if $\lambda_+ < 1 < \lambda_-$, then $\mathscr{D}_z'$ is nonincreasing on $[\lambda_+, \lambda_-]$.*

Proof. We prove the five statements in the order in which they are used.
The endpoint reserve. Differentiation gives

$$R'(z) = 4\log(1+z) - \log(1-2z) - 3\log(1+2z) \ge 0,$$

because $1+z$ is the average of one copy of $1-2z$ and three copies of $1+2z$, and log is concave. Thus $R \ge 0$. Put $t = z/(1+z) \in [0, 1/3]$. A simplification gives

$$2(E-R)' = 8\log(1+t) + \log(1-2t).$$

The right side is concave in t, vanishes at 0, and is positive at $1/3$ because $(4/3)^8 > 3$. Hence $E - R \ge 0$.
The endpoint slope. Using $\operatorname{artanh} r = \sum_{k\ge 0} r^{2k+1}/(2k+1)$ in the derivative of (67) gives

$$\mathscr{D}_z'(1) = 2\sum_{k\ge 1}\frac{4^k-1}{2k(2k+1)}z^{2k}. \tag{72}$$

The even coefficients of $R(z)$ are nonpositive, while its positive odd part is

$$2\sum_{k\ge 1}\frac{2^{2k+1}-2}{2k(2k+1)}z^{2k+1}.$$

Since $z \le 1/2$,

$$(2^{2k+1}-2)z \le 4^k - 1.$$

Comparison term by term proves $\mathscr{D}_z'(1) \ge R(z)$.
The chord. Put $A_0 = 1-z^2$, $B_0 = 1-4z^2$, and

$$\phi(y) = \frac{(1+\sqrt{y})\log(1+\sqrt{y}) - (1-\sqrt{y})\log(1-\sqrt{y})}{2\sqrt{y}}.$$

Direct differentiation gives

$$\mathscr{D}_z'(\lambda) = 2\lambda^{-3}\big[\phi(1-A_0\lambda^4) - \phi(1-B_0\lambda^4)\big]. \tag{73}$$

The difference has the positive integral representation

$$\phi(1-A_0x)-\phi(1-B_0x)=(A_0-B_0)x\int_0^\infty \frac{m(u)}{(u+A_0x)(u+B_0x)}\,du, \tag{74}$$

where

$$m(u)=\frac{\sqrt{1+u}-1}{2\sqrt{1+u}}.$$

To verify (74), differentiate both sides with respect to A_0; after the substitution $r=\sqrt{1-A_0x}$ the resulting scalar integral is immediate, and both sides vanish at $A_0=B_0$. With $x=e^\xi$ and $u=e^{\xi+\zeta}$, the integral in (74) is a convolution of

$$\xi\longmapsto m(e^\xi),\qquad \zeta\longmapsto \frac{e^\zeta}{(e^\zeta+A_0)(e^\zeta+B_0)}.$$

Both functions are log-concave. For the first,

$$\frac{d^2}{d\xi^2}\log m(e^\xi)=-\frac{e^\xi(\sqrt{1+e^\xi}+2)}{4(1+e^\xi)^2}<0;$$

for the second, differentiate $\zeta-\log(e^\zeta+A_0)-\log(e^\zeta+B_0)$. We use the one-dimensional fact that a convolution of log-concave functions is log-concave. Here is a compact proof: after normalization, let T be the increasing transport from f_0 to f_1; for $z=(1-\theta)x+\theta T(x)$, the transport identity and weighted AM–GM give $h(z)z'(x)\ge f_0(x)$, and integration is the Prékopa–Leindler inequality. Applying this with $f_j(y)=f(y)g(x_j-y)$ proves the convolution claim. Hence $\lambda\mapsto\mathscr{D}_z'(\lambda)$ is log-concave in $\log\lambda$.

It follows that every superlevel set of $\mathscr{D}_z'$ is an interval in logarithmic coordinates. Since $\mathscr{D}_z(0)=0$, $\mathscr{D}_z(1)=E$, $\mathscr{D}_z'(1)\ge R$, and

$$\int_0^1(\mathscr{D}_z'(\lambda)-R)\,d\lambda=E-R\ge 0,$$

one has

$$\int_\lambda^1(\mathscr{D}_z'(r)-R)\,dr\ge 0\qquad(0<\lambda\le 1).$$

This is exactly (71).

The hyperbolic weights. Let $y=2\tau$ and set

$$X=\frac{e^y}{\cosh y+z\sinh y},\qquad Y=\frac{e^{-y}}{\cosh y-z\sinh y}.$$

Then $\eta^{-1}\lambda_+=\sqrt{X}$ and $\eta\lambda_-=\sqrt{Y}$. With $T=e^{2y}$,

$$2-X-Y=\frac{2z(1-z)(T-1)^2}{((1-z)T+1+z)((1+z)T+1-z)}\ge 0.$$

Concavity of the square root gives $\sqrt{X}+\sqrt{Y}\le 2$.

The crossing interval. Define, for $0<y\le 1$,

$$F(y)=\frac{\operatorname{artanh}\sqrt{1-y}}{\sqrt{1-y}}+\frac{1}{2}\log y,\qquad H(y)=4yF'(y)-3F(y).$$

A differentiation gives

$$\mathscr{D}_z''(\lambda)=2\lambda^{-4}\big[H(A_0\lambda^4)-H(B_0\lambda^4)\big]. \tag{75}$$

Put $r=\sqrt{1-y}$. Then

$$H'(y)=\frac{N(r)}{2r^5},\qquad N(r)=(6-5r^2)\operatorname{artanh} r-6r+3r^3,$$

and

$$\frac{N(r)}{r^5}=\sum_{m\ge 0}\frac{2m-7}{(2m+3)(2m+5)}r^{2m}. \tag{76}$$

The first four coefficients are negative and all later coefficients are positive. This series has exactly one zero on $(0,1)$: if it vanishes at $x_0 = r_0^2$ and $x > x_0$, subtract $(x/x_0)^3$ times the zero value; every remaining term is positive. Since the series is negative at 0 and tends to $+\infty$ at 1, the function H first increases and then decreases.

Set $y_0 = 128/1089$, so $\sqrt{1-y_0} = 31/33$. Since $H(0) = -3\log 2$,

$$H(y_0) - H(0) = \frac{6705}{29791}\log 2 + 3\log\frac{33}{32} - \frac{256}{961} < 0.$$

Indeed, $\log 2 < 7/10$ (because the first four terms of the exponential series at $7/10$ already exceed 2) and $\log(33/32) < 1/32$, and the resulting rational upper bound is $-14387/953312$.

In the crossing regime $\lambda_+ < 1 < \lambda_-$, put $t = \tanh\tau$. The inequality $\lambda_- > 1$ is equivalent to $0 < t < z$. Hence, throughout $[\lambda_+, \lambda_-]$,

$$A_0\lambda^4 \geq A_0\lambda_+^4 \geq \frac{(1-z^2)^3}{(1+3z^2)^2} \geq \frac{27}{196} > y_0.$$

Because H has one maximum and has already fallen below $H(0)$ by y_0, the minimum of H on $[0, A_0\lambda^4]$ is attained at the right endpoint. Since $0 \leq B_0\lambda^4 \leq A_0\lambda^4$, we get

$$H(A_0\lambda^4) \leq H(B_0\lambda^4).$$

Equation (75) proves $\mathscr{D}_z'' \leq 0$ on the crossing interval. □

7.5.2 Use the scalar toolkit

Proposition 7.9 (The one-small line is nonnegative)**.** *For every q on (64), $G(q) \geq 0$.*

Proof. For $\tau > 0$ one always has $\lambda_+ < 1$, so there are two cases. First suppose $\lambda_- \leq 1$. Apply the chord bound (71) to both λ_+ and λ_- in (69). The terms containing E cancel, and the resulting lower bound is

$$\big[2 - \eta^{-1}\lambda_+ - \eta\lambda_-\big]R,$$

which is nonnegative by the hyperbolic-weight inequality.

Now suppose $\lambda_+ < 1 < \lambda_-$. The crossing monotonicity gives

$$E - \mathscr{D}_z(\lambda_+) \geq (1-\lambda_+)\mathscr{D}_z'(1),$$

$$\mathscr{D}_z(\lambda_-) - E \leq (\lambda_- - 1)\mathscr{D}_z'(1).$$

Rearranging (69), its right side is at least

$$\big[\eta^{-1}(1-\lambda_+) - \eta(\lambda_- - 1)\big]\mathscr{D}_z'(1) - (\eta + \eta^{-1} - 2)R.$$

The hyperbolic-weight inequality makes the bracket at least $\eta + \eta^{-1} - 2$, while the endpoint-slope inequality gives $\mathscr{D}_z'(1) \geq R$. The expression is therefore nonnegative. □

Completion of Theorem 2.11. The KKT reduction and theorem 7.9 give $G \geq 0$ for every even law; (35) restores $d\chi_{123}$, and the endpoint channels follow by continuity. □

8 Proof of the even four-bit theorem

8.1 Klein-four reduction

Let the eight antipodal classes be identified with $\mathbb{F}_2^3$. Set

$$H = \{0,3,5,6\} \cong \mathbb{F}_2^2,$$

so the quotient splits into two cosets $A = H$ and $B = H \oplus 1$. Let T_h be translation by h, let $E_h = (I + T_h)/2$, and let $E_H = |H|^{-1}\sum_{h\in H} T_h$.

For any H-covariant output family and its full twirl, the Pythagorean identity gives

$$4D(\rho_q\|\rho_{E_Hq}) - \sum_{h\neq 0} D(\rho_{E_hq}\|\rho_{E_Hq}) = D(\rho_q\|\rho_{E_Hq}) + \sum_{h\neq 0} D(\rho_q\|\rho_{E_hq}).$$

We prove the repaired Klein inequality

$$4\mathcal{G}_s(q) \geq \mathcal{G}_s(E_Hq) + \sum_{h\neq 0} \mathcal{G}_s(E_hq). \tag{77}$$

Each law on the right depends on at most two disjoint block parities, so Theorem 2.3 makes every term nonnegative. It remains to prove (77).

Resolving global output parity splits both quantum and classical Klein asymmetries into a positive and a negative branch, of traces

$$\pi_\pm = \frac{1 \pm \mu^4}{2}.$$

The next three subsections prove the two branch inequalities.

8.2 The Fano reserve

8.2.1 1. A sharp reverse entropy inequality on two bits

For a probability law p on $\mathbb{F}_2^2$, let u_4 be uniform, and for each nonzero linear form $a \in (\mathbb{F}_2^2)^*$, let p_a be the law of $a(X)$. Put

$$\mathfrak{h}(x) := \frac{(1+x)\log(1+x) + (1-x)\log(1-x)}{2}.$$

Thus, if $\widehat{p}(a) = \mathbb{E}(-1)^{a(X)}$, then

$$D(p_a\|u_2) = \mathfrak{h}(\widehat{p}(a)).$$

Theorem 1 (sharp two-bit reverse entropy inequality) For every law p on $\mathbb{F}_2^2$,

$$\boxed{D(p\|u_4) \leq \kappa_2 \sum_{a\neq 0} D(p_a\|u_2),} \tag{1.1}$$

where

$$\boxed{\kappa_2 = \frac{\log(4/3)}{3\mathfrak{h}(1/3)} = \frac{\log(4/3)}{\log(32/27)} = 1.6932531100704\ldots.} \tag{1.2}$$

The constant is optimal. Apart from affine symmetries, equality holds at the law that is uniform on three of the four points.

Proof Write

$$r_a = \widehat{p}(a), \qquad R = \sum_{a\neq 0} r_a^2, \qquad x_z = 4p(z).$$

Parseval gives

$$\sum_z x_z = 4, \qquad \sum_z x_z^2 = 4(1+R). \tag{1.3}$$

The denominator in (1.1) is

$$A(p) := \sum_{a\neq 0} \mathfrak{h}(r_a).$$

Since $y \mapsto \mathfrak{h}(\sqrt{y})$ is convex,

$$\boxed{A(p) \geq 3\mathfrak{h}\left(\sqrt{R/3}\right).} \tag{1.4}$$

Indeed, with $x = \sqrt{y}$,

$$\frac{d^2}{dy^2}\mathfrak{h}(\sqrt{y}) = \frac{x/(1-x^2) - \operatorname{artanh} x}{4x^3} \geq 0.$$

The range $0 \le R \le 1/3$ For $R < 1/3$, no x_z can vanish: three nonnegative numbers of sum four have squared sum at least $16/3$. At a maximizer of

$$\frac{1}{4}\sum_z x_z \log x_z$$

subject to (1.3), the positive x_z's take at most two values. Put

$$u = \sqrt{3R} \in [0,1].$$

The three multiplicity patterns are

$$(1-u, 1+u/3, 1+u/3, 1+u/3),$$
$$(1+u, 1-u/3, 1-u/3, 1-u/3),$$

and

$$(1+u/\sqrt{3}, 1+u/\sqrt{3}, 1-u/\sqrt{3}, 1-u/\sqrt{3}).$$

The power series

$$(1+x)\log(1+x) = x + \sum_{n\ge 2} \frac{(-1)^n}{n(n-1)} x^n \tag{1.5}$$

shows that the first pattern dominates the second: their difference is

$$2\sum_{k\ge 1} \frac{1-3^{-2k}}{(2k+1)(2k)} u^{2k+1} > 0. \tag{1.6}$$

It also dominates the $2+2$ pattern. Their quadratic terms agree; the first pattern has positive odd terms; and for every $k \ge 2$,

$$1 + 3^{1-2k} \ge \frac{4}{3^k},$$

because

$$3^{2k-1} + 1 - 4\cdot 3^{k-1} = (3^k - 1)(3^{k-1} - 1) \ge 0.$$

Hence

$$D(p\|u_4) \le \frac{1}{4} F(u), \tag{1.7}$$

where

$$F(u) := (1-u)\log(1-u) + 3(1+u/3)\log(1+u/3).$$

The ratio

$$\frac{F(u)}{12\mathfrak{h}(u/3)}$$

is increasing on $[0,1]$. For the even part of F, the coefficient ratios against $12\mathfrak{h}(u/3)$ are

$$\frac{3^{2k-1}+1}{4}, \qquad k \ge 1,$$

which increase with k. After factoring u^3 from the odd part of F and u^2 from the denominator, the coefficient ratios are

$$\frac{(3^{2k+2}-1)(2k+1)}{12(2k+3)}, \qquad k \ge 0,$$

which also increase. The standard power-series ratio lemma therefore applies to both parts. Consequently,

$$\frac{D(p\|u_4)}{A(p)} \le \frac{F(1)}{12\mathfrak{h}(1/3)} = \frac{\log(4/3)}{3\mathfrak{h}(1/3)}. \tag{1.8}$$

The range $1/3 \le R \le 3$ Rényi monotonicity gives

$$D(p\|u_4) \le D_2(p\|u_4) = \log(1+R). \tag{1.9}$$

Put $r = \sqrt{R/3} \in [1/3, 1]$. The function

$$\frac{\log(1+3r^2)}{\mathfrak{h}(r)}$$

is decreasing on this interval. This is the same one-variable monotonicity used in the sharp weak Klein-midpoint theorem: differentiating reduces it to

$$F_0(r) = \log(1+3r^2)\operatorname{artanh} r - \frac{6r}{1+3r^2}\mathfrak{h}(r) > 0,$$

and

$$F_0'(r) = \frac{\log(1+3r^2)}{1-r^2} - \frac{6(1-3r^2)}{(1+3r^2)^2}\mathfrak{h}(r) \ge 0.$$

For $r \ge 1/\sqrt{3}$ the last inequality is immediate. For $1/3 \le r \le 1/\sqrt{3}$, use

$$\mathfrak{h}(r) \le \frac{r^2}{2(1-r^2)}$$

and

$$\log(1+x) \ge \frac{x}{1+x} \ge \frac{x(1-x)}{(1+x)^2}.$$

Finally $F_0(1/3) > 0$, e.g. from the elementary rational bounds

$$\log(4/3) < 47/162, \qquad \log 2 > 56/81.$$

It follows from (1.4) and (1.9) that the ratio in (1.1) is again maximized at $R = 1/3$.

At $R = 1/3$, equality forces

$$(x_z)_z = (0, 4/3, 4/3, 4/3)$$

up to permutation, and the three nontrivial biases have absolute value $1/3$. This is precisely the law uniform on three points. □

8.2.2 2. The sharp Fano-plane hierarchy

Let $Y = (X, C) \in \mathbb{F}_2^3$, and regard the seven nonzero linear forms on Y as the points of the Fano plane. Let c denote the point corresponding to C. For a point p, let H_p be the entropy of the associated bit. For a Fano line L, let H_L be the entropy of the associated rank-two observation.

Define

$$d_p := \log 2 - H_p, \qquad d_L := 2\log 2 - H_L.$$

For the three classical Klein reveal levels from the repaired branch allocation,

$$J_2 = 3\sum_{p \ne c} d_p, \tag{2.1}$$

while

$$J_3 = \sum_{L \not\ni c} d_L - \frac{1}{2}\sum_{p \ne c} d_p. \tag{2.2}$$

Apply Theorem 1 to the rank-two observation on each of the four lines not containing c:

$$d_L \le \kappa_2 \sum_{p \in L} d_p.$$

Every point $p \ne c$ lies on exactly two lines not containing c. Therefore

$$\sum_{L \not\ni c} d_L \le 2\kappa_2 \sum_{p \ne c} d_p.$$

Combining with (2.1)–(2.2) proves:

Theorem 2 (sharp Fano hierarchy)

$$\boxed{J_3 \le \rho_* J_2,} \tag{2.3}$$

where

$$\boxed{\rho_* = \frac{4\kappa_2 - 1}{6} = \frac{\log(8/3)}{6\log(32/27)} = 0.962168740046968\ldots.} \tag{2.4}$$

The constant is optimal. Equality is attained when C is deterministic and X is uniform on three points of H.

8.3 Negative global-parity branch

8.3.1 1. Golden–Thompson overlap entropy

For a positive matrix X, write

$$\mathsf{H}(X) = -\operatorname{Tr}(X \log X).$$

Theorem 1 Let

$$A = \sum_{i=1}^{N} a_i |e_i\rangle\langle e_i|, \qquad B = \sum_{j=1}^{N} b_j |f_j\rangle\langle f_j|$$

be positive semidefinite, and put

$$w_{ij} = |\langle e_i, f_j\rangle|^2.$$

Then

$$\boxed{\mathsf{H}(A+B) \ge -\sum_{i,j} w_{ij}(a_i + b_j)\log(a_i + b_j).} \tag{3}$$

Proof For $z, t > 0$, Golden–Thompson gives

$$\operatorname{Tr} e^{-t(A+B+zI)} \le e^{-tz} \sum_{i,j} w_{ij} e^{-t(a_i+b_j)}.$$

The Frullani representation of log, followed by the scalar identity for $x \log x$, yields the exact reserve formula

$$\begin{aligned}
&\sum_{i,j} w_{ij}(a_i + b_j)\log(a_i + b_j) - \operatorname{Tr}\big((A+B)\log(A+B)\big) \\
&\quad = \int_0^\infty \left[\operatorname{Tr}\log(A + B + zI) - \sum_{i,j} w_{ij} \log(a_i + b_j + z) \right] dz \ge 0.
\end{aligned}$$

Equivalently, it is the integrated Golden–Thompson heat-trace gap

$$\int_0^\infty \frac{\sum_{i,j} w_{ij} e^{-t(a_i+b_j)} - \operatorname{Tr} e^{-t(A+B)}}{t^2}\, dt.$$

□

8.3.2 2. A sharper isoclinic MUB theorem

The linear overlap bound (3) is not yet the right interpolation for the negative branch. The isoclinic geometry gives a sharper pairwise binary experiment.

Let $V, W : \mathbb{C}^d \to \mathbb{C}^{2d}$ be isometries satisfying

$$V^*W = \gamma U, \qquad |U_{ij}|^2 = \frac{1}{d}, \qquad r = \sqrt{1-\gamma^2}.$$

For $a, b \in \mathbb{R}^d_+$, put

$$\rho_r(a,b) = V \operatorname{diag}(a) V^* + W \operatorname{diag}(b) W^*.$$

For scalar masses $x, y \geq 0$, let $\mathsf{s}_r(x,y)$ denote the entropy of the two-state pure-state mixture with masses x, y and trace distance r. Its eigenvalues are

$$\lambda_\pm(x,y;r) = \frac{x+y \pm \sqrt{(x-y)^2 + 4(1-r^2)xy}}{2},$$

so

$$\mathsf{s}_r(x,y) = -\sum_{\pm} \lambda_\pm \log \lambda_\pm.$$

Define

$$\mathsf{P}_r(a,b) = \frac{1}{d} \sum_{i,j} \mathsf{s}_r(a_i, b_j). \tag{4}$$

Theorem 2 (isoclinic binary-pair entropy)

$$\boxed{\mathsf{H}(\rho_r(a,b)) \geq \mathsf{P}_r(a,b).} \tag{5}$$

Proof After a fixed unitary change of coordinates, take

$$V = \begin{pmatrix} I \\ 0 \end{pmatrix}, \qquad W = \begin{pmatrix} \gamma U \\ rI \end{pmatrix}.$$

Writing $A = \operatorname{diag}(a)$ and $B = \operatorname{diag}(b)$, Schur complementation gives

$$\begin{aligned} \det(zI + \rho_r) = {} & \det(zI + r^2 B) \\ & \times \det\left(zI + A + U \operatorname{diag}\left(\frac{(1-r^2) z b_j}{z + r^2 b_j} \right) U^* \right). \end{aligned} \tag{6}$$

Apply the logarithmic form of Theorem 1 to the second determinant. Since $|U_{ij}|^2 = 1/d$,

$$\begin{aligned} \log\det(zI + \rho_r) &\geq \frac{1}{d} \sum_{i,j} \log \left[(z + r^2 b_j) \left(z + a_i + \frac{(1-r^2) z b_j}{z + r^2 b_j} \right) \right] \\ &= \frac{1}{d} \sum_{i,j} \log \left[z^2 + z(a_i + b_j) + r^2 a_i b_j \right]. \end{aligned} \tag{7}$$

The quadratic on the right is

$$(z + \lambda_+(a_i, b_j; r))(z + \lambda_-(a_i, b_j; r)).$$

The two spectral measures in (7) have equal total mass and equal first moment. Integrating the log-determinant inequality as in Theorem 1 proves (5). □

8.3.3 3. Equality on the three Klein midpoints

Identify $H \cong \mathbb{F}_2^2$, and use the four-dimensional Clifford Hadamard

$$U = (H_2 \otimes H_2)\,\mathrm{diag}(1,1,1,-1)(H_2 \otimes H_2).$$

Let T_h be translation by $h \in H$, and

$$E_h a = \frac{1}{2}(a + T_h a), \qquad E_h b = \frac{1}{2}(b + T_h b).$$

If $h \neq 0$, both $E_h a$ and $E_h b$ are invariant under the same translation. In the Schur complement (6), the transformed vector

$$b_j \longmapsto \frac{(1-r^2)z b_j}{z + r^2 b_j}$$

has the same invariance. The two four-dimensional matrices in the second determinant of (6) then commute: in Pauli notation they lie in the commuting algebra generated by Z_k and $Z_k X_h$, where $k \cdot h = 0$. Consequently Golden–Thompson is an equality at every z, and

$$\boxed{\mathsf{H}(\rho_r(E_h a, E_h b)) = \mathsf{P}_r(E_h a, E_h b).} \tag{8}$$

The same equality holds after the full H-twirl.
For a functional $F(a,b)$, write its Klein combination as

$$\mathfrak{K}F = F(\bar{a},\bar{b}) - 4F(a,b) + \sum_{h \neq 0} F(E_h a, E_h b). \tag{9}$$

Equations (5) and (8) give the exact reserve identity

$$\boxed{\mathfrak{K}\mathsf{P}_r - \mathfrak{K}\mathsf{H}(\rho_r) = 4\big[\mathsf{H}(\rho_r(a,b)) - \mathsf{P}_r(a,b)\big] \geq 0.} \tag{10}$$

Thus it remains only to control the scalar pair functional P_r.

8.3.4 4. The one-bit excess is jointly convex

Put $\eta(x) = -x \log x$, and define

$$\mathsf{g}_r(x,y) = \mathsf{s}_r(x,y) - (1-r)\eta(x+y) - r\big[\eta(x) + \eta(y)\big]. \tag{11}$$

Lemma 3 For every $0 \leq r \leq 1$, the function $(x,y) \mapsto \mathsf{g}_r(x,y)$ is jointly convex on $\mathbb{R}_+^2$.

Proof Let $m = x + y$, let $z = (x-y)/m$, and write

$$\mathcal{H}(z) = h\left(\frac{1+z}{2}\right), \qquad v(z) = \sqrt{1 - r^2 + r^2 z^2}.$$

Homogeneity gives

$$\mathsf{g}_r(x,y) = m\big[\mathcal{H}(v(z)) - r\mathcal{H}(z)\big]. \tag{12}$$

The bracket is convex in z. Equivalently,

$$F_r(z) = r\mathcal{H}(z) - \mathcal{H}(v(z))$$

is concave. A direct differentiation gives

$$(1-z^2)F_r''(z) = 1 - r - \frac{1-r^2}{v^2}\left[1 - \frac{1-v^2}{v}\operatorname{artanh} v\right]. \tag{13}$$

Put $w = \sqrt{1-v^2} \le r$. Since $w \operatorname{artanh} v \le v$,

$$\frac{1-r^2}{v^2}\left[1 - \frac{1-v^2}{v}\operatorname{artanh} v\right] \ge (1-r)\frac{1+r}{1+w} \ge 1 - r.$$

Hence $F_r'' \le 0$. Equation (12) is the perspective of a convex one-variable function, so it is jointly convex. □

8.3.5 5. Scalar Klein interpolation

Define

$$\mathsf{G}_r(a,b) = \frac{1}{d}\sum_{i,j} \mathsf{g}_r(a_i, b_j).$$

It is jointly convex and invariant under simultaneous translations of a, b. Therefore

$$\mathsf{G}_r(E_h a, E_h b) \le \mathsf{G}_r(a,b), \qquad \mathsf{G}_r(\bar{a}, \bar{b}) \le \mathsf{G}_r(a,b).$$

It follows that

$$\mathfrak{K}\mathsf{G}_r \le 0.$$

Using (11), and observing that

$$\mathsf{P}_0(a,b) = \frac{1}{d}\sum_{i,j} \eta(a_i + b_j), \qquad \mathsf{P}_1(a,b) = H(a) + H(b),$$

we obtain

$$\boxed{\mathfrak{K}\mathsf{P}_r \le (1-r)\mathfrak{K}\mathsf{P}_0 + r\mathfrak{K}\mathsf{P}_1.} \tag{14}$$

In the Klein notation,

$$\mathfrak{K}\mathsf{P}_0 = J_3, \qquad \mathfrak{K}\mathsf{P}_1 = J_4.$$

Combining (10) and (14) proves

$$\boxed{\frac{Q_-}{\pi_-} \le (1-r)J_3 + rJ_4.}$$

8.3.6 6. Payment by the actual classical branch

The negative classical branch budget is

$$\frac{B_-}{\pi_-} = w_2 J_2 + w_3 J_3 + w_4 J_4, \qquad r = w_4, \qquad 1 - r = w_2 + w_3.$$

The sharp Fano hierarchy proved earlier gives

$$J_3 \le \rho_* J_2 < J_2.$$

Hence

$$\begin{aligned}\frac{B_-}{\pi_-} - \frac{Q_-}{\pi_-} &\geq w_2(J_2 - J_3) \\ &\quad + [(1-r)J_3 + rJ_4 - \mathfrak{RP}_r] \\ &\quad + 4\,[\mathsf{H}(\rho_r(a,b)) - \mathsf{P}_r(a,b)] \geq 0.\end{aligned} \tag{15}$$

Equation (15) displays three separate reserves:

1. the Fano $J_2 - J_3$ reserve;
2. the convex one-bit interpolation reserve;
3. the integrated Golden–Thompson reserve.

The negative branch is therefore complete without numerical certification.

8.4 Positive global-parity branch

8.4.1 1. Result

Let

$$H = \{0, 3, 5, 6\} \cong \mathbb{F}_2^2$$

act by translation on the eight antipodal quotient labels. In the repaired Klein-four proof of the four-bit even Conjecture 1 theorem, the negative output-parity branch has already been completed. This note completes the positive output-parity branch.

Write the quotient labels as two cosets

$$A = H, \qquad B = H \oplus 1.$$

For the positive branch, define

$$\mu = \sqrt{1 - s^2}, \qquad \lambda = \frac{s^4}{2}, \qquad w = \frac{s^3(1-s)}{2},$$

and

$$\pi_+ = \mu^2 + \lambda = \frac{1 + \mu^4}{2}, \qquad c = \frac{\mu(1+\mu^2)}{2}.$$

The unnormalized signal Gram matrix is

$$G_+ = \begin{pmatrix} \lambda I_4 + \mu^2 J_4 & cJ_4 \\ cJ_4 & \lambda I_4 + \mu^2 J_4 \end{pmatrix}. \tag{1}$$

The matched classical positive-branch experiment has:

- a common erasure outcome of mass $e = \mu^2 - 4w$;
- one output of mass w for every cross-coset pair $(i, j) \in A \times B$;
- one label-specific output of mass λ for every input label.

For a prior q, let

$$\mathcal{F}_+(q) = H\left(D_q^{1/2} G_+ D_q^{1/2}\right) - \left[e\,\eta\left(\sum_i q_i\right) + w \sum_{i \in A, j \in B} \eta(q_i + q_j) + \lambda \sum_i \eta(q_i) \right], \tag{2}$$

where $\eta(x) = -x \log x$.

The main theorem is the following.

Theorem 1 — same-coset pair equalization Choose any two labels in the same coset. If P averages their two prior masses and fixes the other six masses, then

$$\boxed{\mathcal{F}_+(q) \geq \mathcal{F}_+(Pq)} \qquad (0 \leq s \leq 1). \tag{3}$$

Consequently the positive Klein branch satisfies its full classical budget for every prior and every channel.
Together with the completed negative branch, this proves the repaired Klein-four inequality

$$4G_s(q) \geq G_s(E_H q) + \sum_{h \in H \setminus \{0\}} G_s(E_h q). \tag{4}$$

Every law on the right side is in the solved two-block-parity family. Therefore:

$$\boxed{\text{Every antipodally even four-bit law satisfies Conjecture 1.}} \tag{5}$$

8.4.2 2. Exact swap testing curve

By symmetry it suffices to average two masses in A. Write

$$q_u = (m+u, m-u, a_2, a_3, b_1, b_2, b_3, b_4), \qquad 0 \leq u \leq m, \tag{6}$$

and let q_0 denote the equalized prior.
The classical entropy gained by equalization is

$$C(q_0) - C(q_u) = \lambda \Psi_m(u) + w \sum_{j=1}^{4} \Psi_{m+b_j}(u), \tag{7}$$

where

$$\Psi_p(u) = (p+u)\log(p+u) + (p-u)\log(p-u) - 2p\log p. \tag{8}$$

The quantum entropy gained by equalization is

$$H(M_{q_0}) - H(M_{q_u}) = D(M_{q_u} \| M_{q_0}).$$

The exact swap-testing identity gives

$$\mathcal{F}_+(q_u) - \mathcal{F}_+(q_0) = \int_m^\infty \frac{2u^2}{v(v^2-u^2)} R(v)\, dv, \tag{9}$$

where

$$R(v) = S_{\mathrm{cl}}(v) - S_Q(v)$$

and

$$S_{\mathrm{cl}}(v) = \lambda (v-m)_+ + w \sum_{j=1}^{4} (v - m - b_j)_+. \tag{10}$$

Order the opposite-coset masses as

$$0 \leq b_1 \leq b_2 \leq b_3 \leq b_4.$$

The classical breakpoints are

$$m, \quad m+b_1, \quad m+b_2, \quad m+b_3, \quad m+b_4. \tag{11}$$

The proof has two parts:

1. $R \geq 0$ through the last breakpoint;
2. the local BKM integral is nonnegative.

A one-crossing lemma then upgrades the local statement to every finite amplitude u.

8.4.3 3. Four breakpoint patterns

At the breakpoint $v = m + b_k$, the extrapolated signed prior is

$$(2m + b_k, -b_k, a_2, a_3, b_1, b_2, b_3, b_4). \tag{12}$$

Put

$$\alpha_1 = 4b_1, \qquad \alpha_2 = 3(b_2 - b_1), \qquad \alpha_3 = 2(b_3 - b_2), \qquad \alpha_4 = b_4 - b_3. \tag{13}$$

Up to additional nonnegative point masses, the signed law at the k-th breakpoint is the convex combination of the first k patterns

$$\begin{aligned} P_1 &= (1/4, -1/4, 0, 0, 1/4, 1/4, 1/4, 1/4), \\ P_2 &= (1/3, -1/3, 0, 0, 0, 1/3, 1/3, 1/3), \\ P_3 &= (1/2, -1/2, 0, 0, 0, 0, 1/2, 1/2), \\ P_4 &= (1, -1, 0, 0, 0, 0, 0, 1), \end{aligned} \tag{14}$$

with coefficients $\alpha_1, \ldots, \alpha_k$.

The corresponding classical budgets are

$$L_1 = \frac{\lambda}{4}, \qquad L_2 = \frac{\lambda + w}{3} = \frac{s^3}{6}, \qquad L_3 = \frac{\lambda}{2} + w = \frac{s^3(2-s)}{4}, \qquad L_4 = \lambda + 3w = \frac{s^3(3-2s)}{2}. \tag{15}$$

The map $A \mapsto \operatorname{Tr} A_-$ is convex. Hence it is enough to check the four patterns.

For $0 < s < 1$, the positive Gram matrix is positive definite. Each pattern has exactly one negative weight, so its signed output has exactly one negative eigenvalue. Evaluating the determinant after shifting by the classical budget gives

$$\det(P_1 + L_1 I) = 0, \tag{16}$$

$$\det(P_2 + L_2 I) = -\frac{s^{12}(s-1)^2(s+1)^3}{7776}\left(s^3 + 2s^2 - 2s - 6\right), \tag{17}$$

$$\det(P_3 + L_3 I) = -\frac{s^9(s-1)^2}{32}\left(3s^2 - 2s - 4\right), \tag{18}$$

and

$$\det(P_4 + L_4 I) = -\frac{3s^6(s-1)^2}{8}\left(s^4 - 5s^3 + 8s^2 - 6\right). \tag{19}$$

The three bracketed polynomials are negative on $[0, 1]$. Their negatives have strictly positive Bernstein coefficient lists

$$(6, 20/3, 20/3, 5), \qquad (4, 5, 3), \qquad (6, 6, 14/3, 13/4, 2).$$

Therefore every breakpoint reserve is nonnegative. Since S_Q is convex in v, R is concave between consecutive breakpoints; hence

$$R(v) \geq 0 \qquad (m \leq v \leq m + b_4). \tag{20}$$

After the last breakpoint the classical curve is affine and R remains concave. The exact tail slopes satisfy

$$S'_Q(\infty) - S'_{\mathrm{cl}}(\infty) = s^2(1-s)^2. \tag{21}$$

Thus R has at most one sign change, from positive to negative.

8.4.4 4. Removing the fixed same-coset background

The two unaffected masses a_2, a_3 contribute the same positive operator to M_{q_u} and M_{q_0}. Addition of a common positive operator is a CPTP image of a flagged direct sum, so

$$D(A+T\|B+T) \leq D(A\|B). \tag{22}$$

It therefore suffices, for the local BKM estimate, to discard those two fixed labels. The reduced experiment contains the varying pair and the four opposite-coset labels.

Let

$$C = \lambda + 2\mu^2 = \frac{(1+\mu^2)^2}{2}. \tag{23}$$

The symmetric and antisymmetric combinations of the two varying signal vectors are orthogonal, with squared norms C and λ. The antisymmetric vector is an eigenvector of the midpoint state, with eigenvalue λm. The tangent couples it only to the symmetric vector.

Every opposite-coset signal has the exact star representation

$$|\psi_j\rangle = \mu|e_+\rangle + \sqrt{\lambda}\,|e_j\rangle, \tag{24}$$

where $e_+, e_1, \ldots, e_4$ are orthonormal. A Schur complement therefore gives the local quantum coefficient

$$I_Q = \int_0^\infty \frac{\lambda C\, dz}{(z+\lambda m)\left(z + Cm + \mu^2 z \sum_{j=1}^4 \frac{b_j}{z+\lambda b_j}\right)}. \tag{25}$$

The classical local coefficient is

$$I_{\mathrm{cl}} = \frac{\lambda}{m} + w\sum_{j=1}^4 \frac{1}{m+b_j}. \tag{26}$$

We now prove $I_Q \leq I_{\mathrm{cl}}$.

8.4.5 5. A scalar background envelope

By homogeneity set $m = 1$, and define

$$x_j = \frac{1}{1+b_j}, \qquad X = \sum_{j=1}^4 x_j, \qquad r = \frac{\mu^2}{\lambda}, \qquad \rho = \frac{r}{1+2r} \in [0, 1/2). \tag{27}$$

After setting $z = \lambda y$, equation (25) becomes

$$I_Q = \lambda F(x_1, \dots, x_4),$$

where

$$F = (1+2r)\int_0^\infty \frac{dy}{(1+y)\left[1+y+2r+r\sum_{j=1}^4 g_y(x_j)\right]}, \tag{28}$$

and

$$g_y(x) = \frac{y(1-x)}{yx+1-x}. \tag{29}$$

For $0 \le y \le 1$,

$$g_y(x) \ge y(1-x). \tag{30}$$

For $y \ge 1$,

$$\frac{\partial^2}{\partial x^2} g_y(x) = \frac{2y^2(y-1)}{(yx+1-x)^3} \ge 0,$$

so Jensen gives

$$\sum_{j=1}^4 g_y(x_j) \ge 4 g_y(X/4). \tag{31}$$

Hence $F \le F_\rho(X)$, where, after the compact change of variables $t = y/(1+y)$,

$$F_\rho(X) = \int_0^1 \frac{dt}{1-2\rho U_X(t)}, \tag{32}$$

with

$$U_X(t) = \begin{cases} \dfrac{X-2}{2}\, t, & 0 \le t \le 1/2, \\ \dfrac{t(X+4t-4)}{4-X+2t(X-2)}, & 1/2 \le t \le 1. \end{cases} \tag{33}$$

For $X = 4$, one has $U_4(t) = t$.

8.4.6 6. The increasing-convex-order lemma

Lemma 2 For every increasing convex function φ on an interval containing $(-\infty, 1]$,

$$\boxed{\int_0^1 \varphi(U_X(t))\, dt \le \left(1 - \frac{X}{4}\right)\varphi(0) + \frac{X}{4}\int_0^1 \varphi(t)\, dt} \tag{34}$$

for every $0 \le X \le 4$.

Proof for $0 \le X \le 2$ Put

$$t_0 = 1 - X/4.$$

Then $U_X(t) \le 0$ for $t \le t_0$. On the remaining interval write

$$t = t_0 + \frac{X}{4}v, \qquad 0 \le v \le 1.$$

A direct simplification gives

$$v - U_X(t) = \frac{v(4-X)(1-v)}{2[4 - X + v(X-2)]} \ge 0. \tag{35}$$

Monotonicity of φ now gives (34).

Proof for $2 \le X \le 4$ Here U_X increases from 0 to 1. It is enough to prove the stop-loss inequalities

$$\int_0^1 (U_X(t) - z)_+ \, dt \le \frac{X}{8}(1-z)^2 \qquad (0 \le z \le 1). \tag{36}$$

For $z = U_X(t)$, define

$$D_X(t) = \frac{X}{8}(1 - U_X(t))^2 - \int_t^1 [U_X(v) - U_X(t)] \, dv. \tag{37}$$

Then

$$D_X'(t) = U_X'(t) \left[(1-t) - \frac{X}{4}(1 - U_X(t)) \right]. \tag{38}$$

On $[0, 1/2]$, the bracket equals

$$\frac{(X-4)[(X+2)t - 2]}{8}. \tag{39}$$

On $[1/2, 1]$, it equals

$$-\frac{(X-4)(t-1)(X+4t-4)}{4[4 - X + 2t(X-2)]}. \tag{40}$$

Thus D_X first increases and then decreases. Its minimum is attained at $t = 0$ or $t = 1$. Clearly $D_X(1) = 0$. At the other endpoint,

$$\begin{aligned} D_X(0) &= \frac{X}{8} - \int_0^1 U_X(t) \, dt \\ &= \frac{(4-X)^2}{16(X-2)^3} \left[X^2 - 4X \log(X/2) - 4 \right] \ge 0. \end{aligned} \tag{41}$$

Indeed, the bracket vanishes at $X = 2$, and its derivative is

$$4 \left[\frac{X}{2} - 1 - \log(X/2) \right] \ge 0.$$

This proves (36), hence (34). □

Apply the lemma to

$$\varphi_\rho(u) = \frac{1}{1 - 2\rho u}.$$

It is increasing and convex. Since $\varphi_\rho(0) = 1$ and $U_4(t) = t$,

$$\boxed{F_\rho(X) \le 1 + \frac{X}{4}\,[F_\rho(4) - 1].} \tag{42}$$

This is the key background-interpolation theorem.

8.4.7 7. The binary endpoint

Put

$$a = \frac{1-s}{s} = \frac{w}{\lambda}.$$

At $X = 4$,

$$F_\rho(4) = \frac{1+2r}{2r}\log(1+2r). \tag{43}$$

The relation

$$r = 2a(a+2)(a+1)^2$$

implies

$$1 + 2r = k^2, \qquad k = 2a^2 + 4a + 1.$$

Therefore

$$F_\rho(4) = \frac{2k^2 \log k}{k^2 - 1}.$$

Define

$$M(a) = 1 + 4a - F_\rho(4).$$

One has $M(0) = 0$, and

$$M'(a) = \frac{P(a) + k\log k}{a^2(a+1)^3(a+2)^2}, \tag{44}$$

where

$$P(a) = 4a^7 + 24a^6 + 52a^5 + 46a^4 + 8a^3 - 10a^2 - 4a.$$

Using

$$\log k \ge \frac{2(k-1)}{k+1},$$

we obtain

$$P(a) + k\log k \ge \frac{2a^4(a+2)^2(2a^3 + 8a^2 + 12a + 7)}{(a+1)^2} \ge 0. \tag{45}$$

Hence

$$F_\rho(4) \le 1 + 4a. \tag{46}$$

Combining (42) and (46),

$$F(x_1, \dots, x_4) \le 1 + aX.$$

Returning to the original scale gives

$$I_Q \le \frac{\lambda}{m} + w \sum_{j=1}^{4} \frac{1}{m + b_j} = I_{\mathrm{cl}}. \tag{47}$$

Thus the local swap Hessian is nonnegative.

8.4.8 8. From the local Hessian to every swap amplitude

The local inequality is

$$\int_m^\infty \frac{2R(v)}{v^3}\, dv \ge 0. \tag{48}$$

As shown in the preceding breakpoint analysis, R has at most one sign change, from positive to negative. For $0 < u < m$, write

$$\frac{2}{v(v^2 - u^2)} = \frac{2}{v^3}\, \theta_u(v), \qquad \theta_u(v) = \frac{v^2}{v^2 - u^2}.$$

The function θ_u is decreasing. If v_* is the sign-change point, then

$$(\theta_u(v) - \theta_u(v_*))R(v) \ge 0$$

on both sides of v_*. Therefore

$$\int_m^\infty \frac{2R(v)}{v(v^2 - u^2)}\, dv \ge \theta_u(v_*) \int_m^\infty \frac{2R(v)}{v^3}\, dv \ge 0. \tag{49}$$

Multiplying by u^2 proves (3) for $u < m$. The boundary $u = m$ follows by continuity.

8.4.9 9. Completion of the positive branch and of the even theorem

Every Klein midpoint $E_h q$ is obtained by averaging disjoint pairs within A and within B. The full twirl $E_H q$ is obtained by repeated same-coset pair averaging. Hence Theorem 1 gives

$$\mathcal{F}_+(q) \ge \mathcal{F}_+(E_h q) \qquad (h \ne 0)$$

and

$$\mathcal{F}_+(q) \ge \mathcal{F}_+(E_H q).$$

Consequently

$$\mathcal{F}_+(E_H q) - 4\mathcal{F}_+(q) + \sum_{h \ne 0} \mathcal{F}_+(E_h q) \le 0. \tag{50}$$

This is exactly the positive-parity Klein branch inequality. The negative branch was proved previously by the Golden–Thompson/MUB flag-loss theorem. Adding the two branches gives

$$4G_s(q) \geq G_s(E_H q) + \sum_{h \neq 0} G_s(E_h q). \tag{51}$$

Each law on the right is a two-block-parity law and already satisfies Conjecture 1. Therefore $G_s(q) \geq 0$ for every antipodally even four-bit law.

Completion of Theorem 2.12. The negative and positive branch inequalities add to (77). Theorem 2.3 makes the four right-hand terms nonnegative. Therefore $\mathcal{G}_s(q) \geq 0$ for every antipodally even four-bit law. The endpoint channels follow by continuity. □